

Matching with Bounded Probabilities: Flexibility, Efficiency, and Crisis Dynamics

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Abstract

We propose the Soft-Min Cobb-Douglas (SMCD) matching function, a tractable three-parameter specification that is freely calibratable to any tightness, job-finding rate, and elasticity like Cobb-Douglas, yet keeps matching probabilities bounded like Den Haan–Ramey–Watson (HRW)—resolving a dilemma the two standard benchmarks pose. Its parameters carry a transparent regime interpretation, and a two-plateau variant adds one parameter for regime-specific curvature. On U.S. data (1951–2025; October 2008 break), SMCD dominates HRW while respecting the bounds Cobb-Douglas violates in 9–12% of months, and the two-plateau variant matches Cobb-Douglas wherever the latter is a valid probability model. We embed the estimated matching function in a DMP economy under a split identification—its shape fixed externally by the data, the remaining structural parameters estimated on separate aggregate moments—so the general-equilibrium results reflect matching-function shape, not calibration. Through a modified Hosios condition with fixed matching costs, the shape produces a sign-reversing unemployment gap: the market runs inefficiently tight on average, yet the constant bargaining weight generates large excess unemployment in crises—an inefficiency HRW cannot produce. The near-Leontief recession branch further makes unemployment an order of magnitude more right-skewed than under HRW.

Keywords: matching function, probability bounds, unemployment rate gap, unemployment crises, structural break, simulated method of moments

JEL Classification: J64, E24, E32, C13, C22

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1. Introduction

The matching function is a fundamental building block of equilibrium unemployment models in the tradition of [Diamond \(1982\)](#), [Mortensen and Pissarides \(1994\)](#), and [Pissarides \(1985\)](#). It maps unemployment u and vacancies v into a flow of new matches $m(u, v)$, and its properties shape the steady-state and cyclical behavior of any model in which job creation and destruction are mediated by search frictions. Quantitative work typically summarizes it by two numbers at a calibration point: the *level* of matching, captured by a job-finding probability f^* , and the *elasticity* of matching with respect to vacancies, ε^* , which governs how sensitively hiring responds to labor market tightness $\theta = v/u$. Yet the object that matters most for the cyclical behavior of unemployment is not the elasticity at a point but the *shape* of the function—how that elasticity varies as tightness moves over the business cycle. We show that this shape has first-order consequences that the standard Cobb-Douglas workhorse cannot even represent: it governs both the tail risk of unemployment crises and whether the search equilibrium is constrained-efficient, and it can make a labor market that is inefficiently *tight* on average the source of large excess unemployment in deep recessions.

In practice, a large body of quantitative work relies on the Cobb-Douglas (CD) matching function, $m(u, v) = Au^\alpha v^{1-\alpha}$, because it is analytically convenient and empirical surveys find it fits aggregate data reasonably well in log-linear space ([Petrongolo and Pissarides, 2001](#)). Its appeal for calibration is also clear: given targets $(f^*, \varepsilon^*, \theta^*)$, the parameters $\alpha = 1 - \varepsilon^*$ and $A = f^*/(\theta^*)^{\varepsilon^*}$ follow immediately from closed-form formulas. Yet the Cobb-Douglas function has a basic deficiency: it does not bound the implied matching probabilities. The job-finding probability $f(\theta) = A\theta^{1-\alpha}$ grows without bound as θ increases, violating $f \leq 1$, and the vacancy-filling probability $q(\theta) = A\theta^{-\alpha}$ fails to respect $q \leq 1$ at low tightness. These violations are not a theoretical curiosity. As we document in [Section 3](#), monthly market tightness falls below the lower admissible bound in nearly one in ten months over our 1951–2025 sample, concentrated in every major recession. It is the vacancy-filling bound that fails empirically, not the job-finding bound.

A smooth alternative that satisfies the bounds is the matching function of [den Haan et al. \(2000\)](#), henceforth HRW: $m(u, v) = uv / (u^\nu + v^\nu)^{1/\nu}$. The HRW function is smooth, has constant returns to scale, and bounds both f and q in $(0, 1)$ for all θ . However, it introduces a different limitation: the job-finding elasticity $\varepsilon(\theta) = 1/(1 + \theta^\nu)$ is pinned entirely by the single parameter ν . Given a calibration point θ^* , the researcher can match either the target elasticity ε^* or the target job-finding rate f^* , but not both simultaneously. In practice, [den Haan et al. \(2000\)](#) calibrate ν to match business-cycle moments rather than steady-state targets, acknowledging that joint calibration to level and elasticity is infeasible within this specification. These two functions are the standard benchmarks in the quantitative DMP literature ([Petrosky-Nadeau and Wasmer, 2017](#)), and between them they pose a dilemma: Cobb-Douglas is freely calibratable but unbounded, whereas HRW is bounded but too rigid to match level and elasticity jointly.¹

This paper proposes a matching function that resolves both limitations. We refer to it as the *Soft-Min Cobb-Douglas* (SMCD) matching function:

$$m(u, v) = \frac{sauv}{[(av)^p + u^p]^{1/p}},$$

with parameters $s \in (0, 1)$, $a > 0$, and $p > 0$. We show that it satisfies four desiderata simultaneously: (i) constant returns to scale; (ii) smoothness and concavity; (iii) exact joint calibration to any target triple $(\theta^*, f^*, \varepsilon^*)$, unlike HRW; and (iv) bounded matching probabilities $f, q \in (0, 1)$ for all θ , unlike CD. In short, the SMCD resolves the dilemma: it is freely calibratable like Cobb-Douglas yet bounded like HRW, which it nests as the special case $s = a = 1$, and near the calibration point it coincides with Cobb-Douglas to first order.

The name *soft-min* arises from the fact that the job-finding rate $f(\theta)$ behaves as a smooth minimum of two limiting regimes: a vacancy-limited regime at low tightness where

¹The structural microfoundations of this class of matching functions have been developed by [Stevens \(2007\)](#) and [Angelis and Bramoullé \(2026\)](#); the present paper addresses the distinct question of which reduced-form specification best fits aggregate data while respecting probability constraints that bind in recessions.

$f(\theta) \approx sa\theta$ and each vacancy is filled with near certainty, and a worker-limited regime at high tightness where $f(\theta) \approx s$ and workers face a capacity constraint on successful contacts. The parameter p governs how sharply the transition occurs between regimes. As $p \rightarrow \infty$, the function converges to the hard-min $\min\{sa\theta, s\}$.

We take this specification to U.S. labor market data spanning 1951–2025, with the aim of letting the data discipline the *shape* of the matching function. Using the job-finding rate constructed by the method of [Shimer \(2012\)](#) and a composite vacancy series that splices Help-Wanted Index data with JOLTS openings ([Barnichon, 2010](#)), we estimate SMCD, HRW, and CD and compare their fit. A QLR sup-Wald test with a moving-block bootstrap identifies a structural break in the log-linear relationship between f and θ in October 2008, coinciding with the Lehman Brothers collapse and the onset of the Great Recession. The break reflects both a level shift and a slope change, consistent with the persistent decline in matching efficiency documented by [Barnichon and Figura \(2015\)](#) and [Şahin et al. \(2014\)](#). We let curvature differ between the vacancy-limited and worker-limited regimes through a two-plateau extension that adds a single parameter. Estimated separately on the pre- and post-break subsamples, the two-plateau SMCD attains fit comparable to CD within the latter’s admissible tightness range while always respecting the probability bounds CD frequently violates: the large majority of CD’s unconditional RMSE advantage—83 percent pre-break and 92 percent post-break—comes from observations in which it predicts a vacancy-filling probability exceeding one.

The payoff of a data-disciplined shape appears once we embed it in general equilibrium. We place the pre-break estimates in a calibrated Diamond–Mortensen–Pissarides economy and compare the two-plateau SMCD with HRW under a deliberately split identification. The shape of the matching function is fixed *externally*, by the horse race, from the data on tightness and job finding rates. The remaining structural parameters—the value of non-work b , the bargaining weight α_L , and the vacancy cost k —are then estimated by simulated method of moments on the globally solved ergodic distribution, targeting a separate, non-

overlapping set of aggregate moments: the relative volatility $\sigma(u)/\sigma(z)$, mean tightness $\bar{\theta}$, and the wage elasticity $\varepsilon_{w,y}$. Because these moments discipline levels, volatilities, and wage cyclicalities rather than the curvature of matching, the matching-function shape is never a free parameter tuned to fit the business cycle. Any difference across specifications—and the efficiency and crisis results that follow—is therefore a consequence of a data-disciplined matching technology, not an artifact of the calibration.

To partially address the unemployment-volatility puzzle (Shimer, 2005), we adopt the two-part hiring cost of Pissarides (2009): a fixed matching cost κ paid upon formation plus a flow vacancy cost k . This cost structure implies a *modified Hosios condition* (Proposition 3): the efficient bargaining weight becomes $\alpha_L^H(\theta) = (1 - \phi)\eta_u(\theta)/[1 - \phi\eta_u(\theta)]$, where ϕ is the fixed-cost share. The condition generalizes the standard $\alpha_L = \eta_u$ by recognizing that only the flow component creates a congestion externality. Under Cobb-Douglas, η_u is constant, so the modification is state-dependent only through $\phi(\theta)$ —a mild variation. Under a bounded matching function whose elasticity $\eta_u(\theta)$ varies with the cycle, the condition acquires quantitative bite: in recessions, $\eta_u(\theta) \rightarrow 0$ drives $\alpha_L^H \rightarrow 0$, and a constant bargaining weight that is nearly efficient on average becomes grossly excessive.

Two results follow. First, and more importantly, the state-dependent Hosios condition produces a *sign-reversing unemployment gap*. On average the decentralized 2P-SMCD economy is inefficiently *tight*—firms over-hire, so unemployment sits below its efficient level, consistent with the efficient-unemployment-rate evidence of Michaillat and Saez (2024). However, in crises, the matching elasticity collapses and the constant bargaining weight overshoots the vanishing efficient share, and the gap flips to large *excess* unemployment. HRW, whose calibrated weight sits near the modified Hosios boundary throughout, exhibits no such sign flip. Second, and reinforcing the first, the near-Leontief recession branch generates vacancy shutdowns and fat-tailed crisis dynamics: the simulated unemployment distribution is an order of magnitude more right-skewed under the two-plateau SMCD than under HRW (skewness 13.8 versus 1.8). This is the Petrosky-Nadeau and Zhang (2017, 2021) crisis chan-

nel, which we show survives and sharpens under a matching function *jointly calibrated* to level and elasticity targets. Both results emerge only from the global solution of a model with a bounded, cyclically-varying matching elasticity.

The paper proceeds as follows. Section 2 reviews the related literature. Section 3 describes the data and documents the CD bound violations. Section 4 develops the SMCD matching function and proves its properties, with a numerical example calibrated to the sample means from Section 3. Section 5 presents the empirical horse race. Section 6 embeds the estimated matching functions in a calibrated DMP model and shows that the shape of the matching function produces a state-dependent efficiency wedge with a sign-reversing unemployment gap and governs the tail risk of unemployment crises. Section 7 addresses the objection that probability bounds are irrelevant in continuous-time or high-frequency formulations. Section 8 concludes. The appendices provide additional figures, proofs and derivations, details of the global solution method and estimation, and robustness checks.

2. Related Literature

Petrongolo and Pissarides (2001) provide the canonical survey of matching function estimation. They find that constant returns to scale is broadly consistent with the data and that CD fits aggregate time-series data reasonably well, with estimated unemployment elasticities in the range $[0.5, 0.7]$. Their survey, however, focuses on the interior of the data and does not address the behavior of CD at extreme tightness values where probability bounds are violated. Our paper takes this limitation as its starting point.

The HRW matching function, introduced by den Haan et al. (2000), bounds matching probabilities by construction and has been adopted by Hagedorn and Manovskii (2008) and Petrosky-Nadeau and Zhang (2017), who show that an accurate global solution of the DMP model is critical for capturing the nonlinear dynamics that arise as matching probabilities approach their bounds in recessions. However, HRW's single-parameter structure prevents joint calibration to both level and elasticity targets. The SMCD nests HRW as a special

case ($s = a = 1$) while adding the flexibility needed for joint calibration.² The sensitivity of quantitative predictions to matching function parameters in [Shimer \(2005\)](#), [Hagedorn and Manovskii \(2008\)](#), and [Petrosky-Nadeau and Zhang \(2017\)](#) underscores the importance of being able to hit calibration targets exactly—the joint target that HRW cannot accommodate but the SMCD can.³

Other approaches include [Anderson and Burgess \(2000\)](#), who reject CD’s homogeneity restrictions, and [Borowczyk-Martins et al. \(2013\)](#), who decompose matching into search intensity and contact technology. [Stevens \(2007\)](#) and [Angelis and Bramoullé \(2026\)](#) develop structural microfoundations for the CES family.⁴ [Lange and Papageorgiou \(2020\)](#) nonparametrically estimate the matching function and find that vacancy elasticity is higher at low tightness—consistent with the countercyclical elasticity that probability bounds mechanically require.

The empirical literature has documented substantial time variation in U.S. matching efficiency. [Barnichon and Figura \(2015\)](#) find a sharp decline beginning in 2008–2009, which they attribute to compositional shifts and mismatch; [Şahin et al. \(2014\)](#) estimate that mismatch accounts for a substantial fraction of the post-2008 unemployment increase. Our structural break finding at October 2008 aligns with this evidence.

A separate strand studies the *efficiency* of the search equilibrium. [Michaillat and Saez \(2021, 2024\)](#) develop a sufficient-statistic approach that measures the unemployment gap $u - u^*$ —and the full-employment rate $u^* = \sqrt{uv}$ —from the elasticity of the Beveridge curve and the relative social costs of unemployment and vacancies, without committing to a par-

²The urn-ball matching function, $q(\theta) = (1 - e^{-1/\theta})$, $f(\theta) = \theta q(\theta)$, shares the same one-parameter limitation and is strictly less flexible than HRW in shape. A scaled HRW, $f(\theta) = A\theta/(1 + \theta^\nu)^{1/\nu}$ with $A \leq 1$, would allow joint calibration—but it is literally the SMCD at $s = A$, $a = 1$; adding an asymmetric weighting parameter recovers the full SMCD parameterization. The $s = a = 1$ restriction is the maximal HRW that respects probability bounds without being a relabeled SMCD.

³The same functional-form requirements arise in New Monetarist models of monetary exchange ([Kiyotaki and Wright, 1993](#); [Lagos and Wright, 2005](#)) and in goods-market models of capacity utilization ([Bai et al., 2025](#); [Silva and Urias, 2025](#)), where meeting probabilities must satisfy the same $(0, 1)$ bounds. The SMCD is equally applicable in these settings.

⁴[Angelis and Bramoullé \(2026\)](#) show that a dense-network microfoundation requires $\nu \leq 1$; HRW’s original $\hat{\nu} = 1.27$ falls outside this range.

ticular matching function or wage-setting protocol. [Landais et al. \(2018\)](#) build on the same framework to study optimal unemployment insurance. That model-free robustness rests on an isoelastic Beveridge curve and symmetric recruiting costs. Section 6 takes a complementary, structural route. We embed the two-plateau matching function in an equilibrium unemployment model with two-part hiring costs ([Pissarides, 2009](#)), characterize efficiency through a modified Hosios condition, and trace the resulting unemployment gap over the cycle.

3. Data

3.1. Sources and construction

We use monthly U.S. data from January 1951 through December 2025. The unemployment rate is the civilian unemployment rate (UNRATE) from FRED, expressed as a fraction of the labor force. Vacancy data splice two series. The [Barnichon \(2010\)](#) composite Help-Wanted Index (HWI), which adjusts newspaper help-wanted counts for the secular shift to online job postings and is expressed as vacancies per labor force, runs from 1951 through 2019. JOLTS job openings (JTSJOL), normalized by the civilian labor force (CLF16OV), begin in December 2000. Both series are available over the December 2000 – December 2019 overlap window; we use this overlap only to estimate the level rescaling of the HWI onto the JOLTS scale via a log-mean ratio, and then use JOLTS itself from December 2000 onward. The spliced series is thus the scaled HWI through November 2000 and JOLTS thereafter—using the direct vacancy measure wherever it exists, with the scaling serving only to make the pre-2001 HWI level-consistent with it. Because the rescaling is a single multiplicative constant, the splice introduces no slope change at the seam, and the tightness values near the Cobb-Douglas bound $\theta_{\min}$ arise from recessions spread across the whole sample rather than from the 2000–2001 join. Market tightness is $\theta_t = v_t/u_t$. Monthly job-finding and separation rates follow the continuous-time construction of [Shimer \(2012\)](#), using unemployment levels (UNEMPLOY), employment levels (CE16OV), and short-term unemployment (UEMPLT5).

3.2. Summary statistics and time series

Figure 1 plots the unemployment rate and vacancy rate (left panel) and market tightness (right panel) over the full sample. Periods in which $\theta < \theta_{\text{low}} \approx 0.32$ — where the Cobb-Douglas matching function implies $q > 1$ — are marked in red. These violations are confined to recessions and account for 9.9% of all months in the sample.

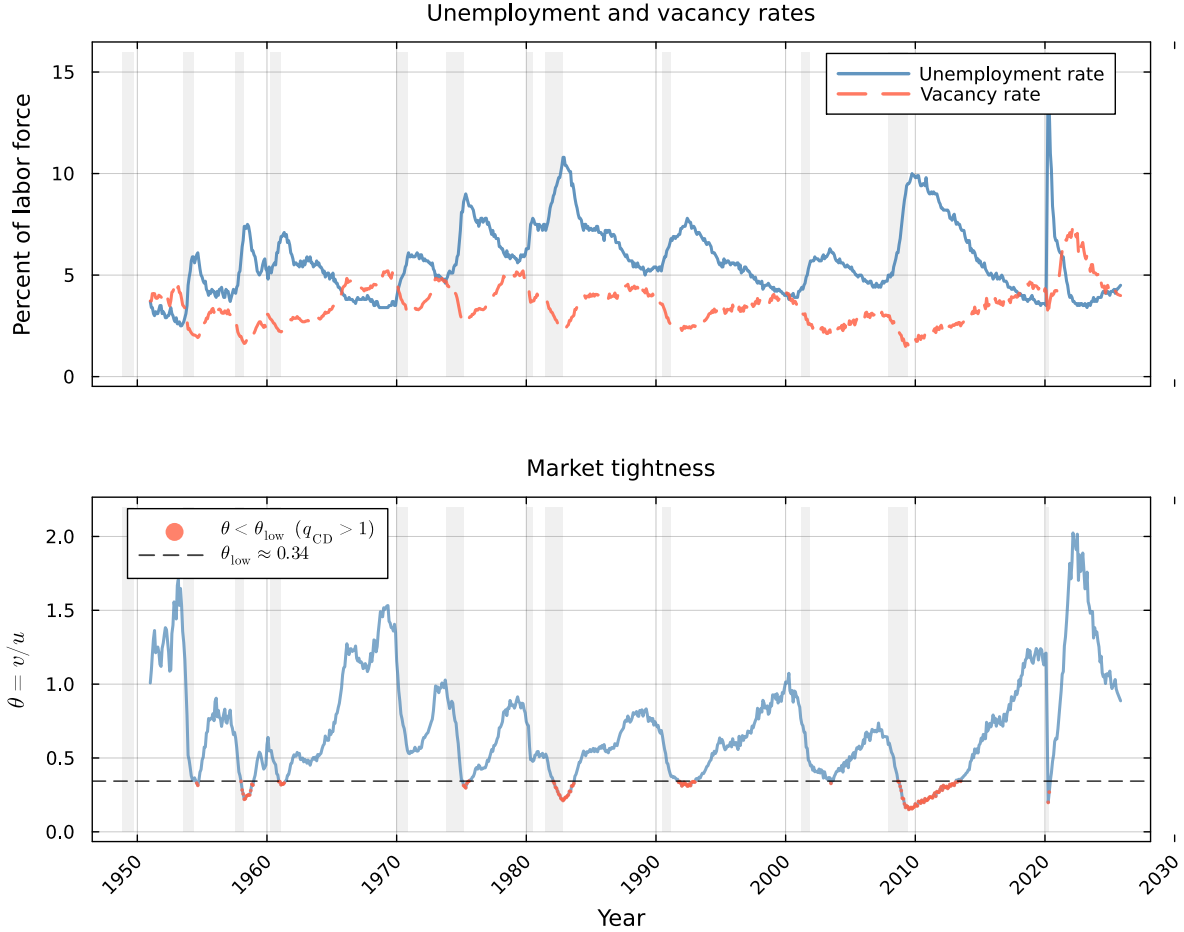


Figure 1: Monthly U.S. labor market data, 1951–2025. Top: unemployment rate and vacancy rate (Barnichon 2010 composite HWI spliced with JOLTS). Bottom: market tightness $\theta = v/u$; red dots mark months with $\theta < \theta_{\text{low}}$ (dashed line), where the Cobb–Douglas matching function implies $q > 1$. Grey bands are NBER recession dates.

Table 1 reports sample means over the full period.⁵

⁵The sample mean of $\theta_t = v_t/u_t$ (0.70) exceeds the ratio of sample means $\bar{v}/\bar{u} = 0.61$ due to Jensen’s inequality and the positive covariance between vacancies and the inverse of unemployment over the cycle. We use $\bar{\theta} = 0.70$ as the calibration target, consistent with treating θ^* as the mean of the tightness series

Table 1: Sample means, U.S. monthly data 1951–2025.

Variable	Mean	Std. dev.	Min	Max
Unemployment rate u	0.057	0.017	0.025	0.148
Vacancy rate v	0.035	0.010	0.014	0.075
Market tightness $\theta = v/u$	0.700	0.358	0.152	2.024
Job-finding rate f	0.402	0.096	0.162	0.691

Note: Monthly, Jan 1951 – Dec 2025. $n = 896$. All rates are fractions of the labor force. Vacancy series splices Barnichon (2010) HWI with JOLTS.

3.3. Preliminary evidence on the matching function

The matching function $m(u, v)$ maps unemployment u and vacancies v into a flow of new matches. Under constant returns to scale, the job-finding probability depends only on market tightness $\theta = v/u$:

$$f(\theta) \equiv \frac{m(u, v)}{u} = m(1, \theta).$$

The matching balance condition $m_t = u_t f_t = v_t q_t$ then pins down the vacancy-filling probability as $q_t = f_t / \theta_t$.

We measure f_t following [Shimer \(2012\)](#). Let $u_t^{<5}$ denote the stock of workers unemployed for less than five weeks (BLS series UEMPLT5), which proxies the flow of new job-losers entering unemployment during month t . Write next-period unemployment as $u_{t+1} = u_t(1 - f_t) + u_{t+1}^{<5}$ and rearrange for the job finding rate $f_t = 1 - \frac{u_{t+1} - u_{t+1}^{<5}}{u_t}$. Estimating the Cobb-Douglas matching function by OLS in log-log space on the full sample yields $\log f_t = \log A + \varepsilon \log \theta_t + e_t$, with $\hat{A} \approx 0.45$ and $\hat{\varepsilon} \approx 0.29$ ($\hat{\alpha} = 1 - \hat{\varepsilon} \approx 0.71$), consistent with [Shimer \(2005\)](#) and within the range surveyed by [Petrungolo and Pissarides \(2001\)](#). These estimates imply the CD probability bounds hold only for $\theta \in [0.32, 15.7]$, with 9.9% of monthly observations violating the lower bound. Section 5 examines whether SMCD fits these data better than

rather than the ratio of its component means.

CD or HRW, and whether the relationship between f and θ is stable over the sample.

4. The Soft-Min Cobb-Douglas Matching Function

We propose the Soft-Min Cobb-Douglas (SMCD) matching function, defined as

$$m(u, v) = \frac{sa uv}{[(av)^p + u^p]^{1/p}}$$

with parameters $s \in (0, 1), a > 0, p > 0$. The denominator is the L^p norm of the vector (u, av) . The HRW matching function is a special case with $s = a = 1$. The matching probabilities are

$$\begin{aligned} f(\theta) &= \frac{m}{u} = \frac{sa\theta}{[(a\theta)^p + 1]^{1/p}} = \frac{s}{(1 + (a\theta)^{-p})^{1/p}} \\ q(\theta) &= \frac{m}{v} = \frac{sa}{[(a\theta)^p + 1]^{1/p}} \end{aligned} \tag{1}$$

The elasticity of the job finding rate is

$$\varepsilon(\theta) = \frac{(a\theta)^{-p}}{1 + (a\theta)^{-p}} = \frac{1}{(a\theta)^p + 1} \tag{2}$$

Note that $f(\theta) \leq s$ and $q(\theta) \leq sa$ for all θ . Moreover, $\varepsilon(0) = 1$ and decreases monotonically to zero as $\theta \rightarrow \infty$. This countercyclical pattern is mechanically implied by the bound: since $f(\theta) \leq s$, any bounded increasing job-finding rate must saturate as $\theta \rightarrow \infty$, driving $\varepsilon(\theta) \rightarrow 0$ in tight markets. The point $a\theta = 1$ characterizes the cross-over at which $\varepsilon = 1/2$.

Constant returns to scale is immediate: $m(\lambda u, \lambda v) = \lambda^2 sa uv / \{\lambda [(av)^p + u^p]^{1/p}\} = \lambda m(u, v)$.

Lemma 1. *Suppose $m(u, v)$ has constant returns to scale. Then m is concave if and only if $f(\theta) \equiv m(1, \theta)$ is concave.*

The proof is standard: sufficiency follows from the restriction of a concave function to an affine set; necessity follows from the convex-combination decomposition $m(tu_1 + (1 -$

$t)u_2, tv_1 + (1 - t)v_2) \geq t m(u_1, v_1) + (1 - t) m(u_2, v_2)$ using the weights $\lambda_i = tu_i/u'$. It therefore suffices to verify concavity of f .

Lemma 2. *The job finding probability $f(\theta) = sa\theta[1 + (a\theta)^p]^{-1/p}$ is concave for all $a, p > 0$.*

Proof. See [Appendix B.1](#). □

Let us check (iii) (joint calibration). That is, given θ^* , we require the matching function to satisfy any $f^* = f(\theta^*) \leq \theta^*$ and elasticity ε^* . First, given p , we characterize choices of a and s which satisfy the job-finding rate and elasticity. Then we find a value of p such that the bounds on the matching probabilities are satisfied. As an intermediate step, we write the job-finding rate in terms of the elasticity using (1) and (2):

$$f(\theta) = s(1 - \varepsilon(\theta))^{1/p}$$

This interpretation emphasizes the fact that $f(\theta)$ is bounded above by s , and that the deviation depends on the elasticity scaled by $1/p$. Given p , choosing $a = (\theta^*)^{-1}[1/\varepsilon^* - 1]^{1/p}$ and $s = f^*/(1 - \varepsilon^*)^{1/p}$ ensures $\varepsilon(\theta^*) = \varepsilon^*$ and $f(\theta^*) = f^*$. To bound the matching probabilities, we need p such that $s \leq 1$ and $sa \leq 1$:

$$s = f^*/(1 - \varepsilon^*)^{1/p} \leq 1, \quad sa = (f^*/\theta^*)(\varepsilon^*)^{-1/p} \leq 1 \quad (3)$$

We can always satisfy these requirements for p sufficiently large since for $p \rightarrow \infty$ these requirements reduce to simply $f^* \leq 1$ and $f^* \leq \theta^*$, which is simply a requirement that the approximation target make sense. Specifically, it follows from (3) that it is sufficient to take $p \geq p_{\min}$ where

$$p_{\min} = \max\{p_s, p_{sa}\}, \quad p_s = \log(1 - \varepsilon^*)/\log(f^*), \quad p_{sa} = \log(\varepsilon^*)/\log(f^*/\theta^*)$$

We thus conclude that the SMCD matching function satisfies all properties (i)–(iv).

The SMCD matching function is isomorphic to a CES matching function $m(u, v) = A[\eta u^\rho + (1 - \eta) v^\rho]^{1/\rho}$. Applying the identity $xy/(x^p + y^p)^{1/p} = (x^{-p} + y^{-p})^{-1/p}$ with $x = u$, $y = av$ rewrites SMCD as $m = s[u^{-p} + a^{-p}v^{-p}]^{-1/p}$, which matches the CES form under

$$\rho = -p, \quad \eta = \frac{a^p}{1 + a^p}, \quad A = \frac{sa}{(1 + a^p)^{1/p}},$$

with inverse $p = -\rho$, $a = [\eta/(1 - \eta)]^{-1/\rho}$, $s = A\eta^{1/\rho}$.

Despite this formal equivalence, the SMCD parameterization is preferable for three reasons. First, probability-bound satisfaction requires $\rho < 0$ in the CES — only the negative CES subfamily generates $f, q \in (0, 1)$ for all $\theta > 0$, a non-obvious restriction that the SMCD renders explicit as $p \geq p_{\min}$. Second, given any $\rho < 0$, both parameterizations admit closed-form solutions for the remaining parameters from a calibration triple $(f^*, \varepsilon^*, \theta^*)$. The difference is that the SMCD bound conditions $s \leq 1$ and $sa \leq 1$ are immediately legible from (1) and yield $p_{\min}$ in two lines, whereas the equivalent CES constraints $A \leq \eta^{-1/\rho}$ and $A \leq (1 - \eta)^{-1/\rho}$ are opaque joint restrictions that give no guidance on how to choose ρ . Third, the CES share $\eta = a^p/(1 + a^p)$ conflates the crossover tightness and the curvature without a direct economic interpretation. By contrast, the SMCD parameters s , a , and p each carry a transparent meaning (upper bound, inverse crossover, bottleneck sharpness). HRW is itself CES with $\eta = \frac{1}{2}$, $\rho = -\nu$, $A = 2^{-1/\nu}$, but features a more transparent parameterization. SMCD relaxes HRW's symmetry restriction ($s = a = 1$) in the minimal way needed for joint calibration to level and elasticity targets.

The job-finding rate can be rewritten as

$$f(\theta) = \frac{1}{(s^{-p} + (sa\theta)^{-p})^{1/p}},$$

the generalized soft-min of two caps: a worker-limited ceiling s and a vacancy-limited ceiling $sa\theta$. As $p \rightarrow \infty$, $f \rightarrow \min\{sa\theta, s\}$; at $p = 1$ it reduces to the harmonic mean $sa\theta/(1 + a\theta)$, which at $s = a = 1$ is the [Kiyotaki and Wright \(1993\)](#) bilateral random meeting function.

At low tightness the vacancy-limited regime binds ($f \approx sa\theta$, $\varepsilon \approx 1$); at high tightness the worker-limited regime binds ($f \approx s$, $\varepsilon \approx 0$). The crossover $\theta = 1/a$ marks $\varepsilon = 1/2$; p governs the sharpness of the transition. Figure 2 illustrates both regimes at the baseline calibration $(\theta^*, f^*, \varepsilon^*) = (0.70, 0.40, 0.29)$ with $p = p_{\min}$.

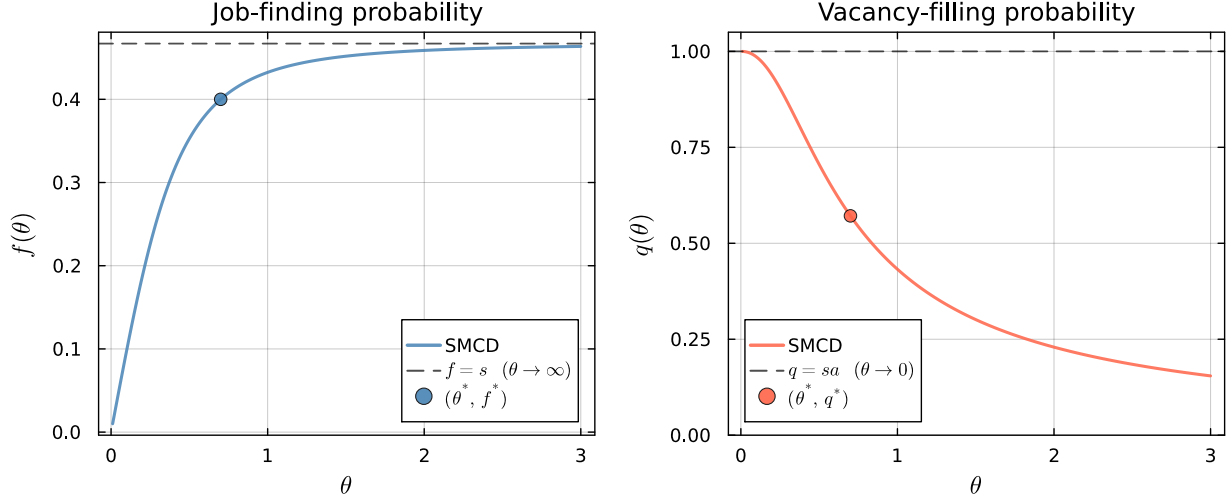


Figure 2: Job-finding probability $f(\theta)$ (left) and vacancy-filling probability $q(\theta)$ (right) under SMCD. Calibration: $(\theta^*, f^*, \varepsilon^*) = (0.70, 0.40, 0.29)$, $p = p_{\min} \approx 2.21$. Dashed lines: asymptotic bounds $f = s$ and $q = sa$. Filled circle: calibration point (θ^*, f^*) .

Local Cobb-Douglas behavior is a corollary of (ii) and (iii): smoothness and exact calibration give $d \log m = (1 - \varepsilon^*)d \log u + \varepsilon^* d \log v$ at θ^* . As a numerical illustration, we calibrate to U.S. sample means $(\theta^*, f^*, \varepsilon^*) = (0.70, 0.40, 0.29)$ from Section 3. These targets imply $p_{\min} \approx 2.21$, with the vacancy-filling constraint $sa \leq 1$ binding. Setting $p = p_{\min}$ yields $s \approx 0.467$ and $a \approx 2.14$, so that $sa \approx 1.000$.

At this calibration, CD implies $q > 1$ for $\theta < 0.32$ — 9.9% of monthly observations (Section 3). HRW respects both bounds but its elasticity at θ^* differs from ε^* by construction; no other specification in the quantitative DMP literature satisfies all four properties simultaneously.

The bound constraint pins p only from below: any $p \geq p_{\min}$ is feasible. Since CD has zero curvature in log space, the SMCD closest to CD minimizes $|\kappa(\theta^*)| = p\varepsilon^*(1 - \varepsilon^*)$, which is increasing in p .

Proposition 1. *The SMCD that is closest to Cobb-Douglas in the neighborhood of θ^* , subject to $p \geq p_{\min}$, has $p = p_{\min}$.⁶*

As p rises from $p_{\min}$ toward infinity, the job-finding rate approaches the Leontief profile $\min\{sa\theta, s\}$ (Figure A.2 in the appendix). Figure 3 plots the matching elasticity $\varepsilon(\theta)$ and its log-curvature $\kappa(\theta) = -p\varepsilon(1-\varepsilon)$ at $p = p_{\min}$. The elasticity is strictly decreasing in θ , passing through $\varepsilon^* = 0.29$ at θ^* and through $1/2$ at the crossover $\theta = 1/a \approx 0.47$; $|\kappa|$ is maximized at the crossover. This non-constant elasticity distinguishes SMCD from Cobb-Douglas and aligns with Lange and Papageorgiou (2020).

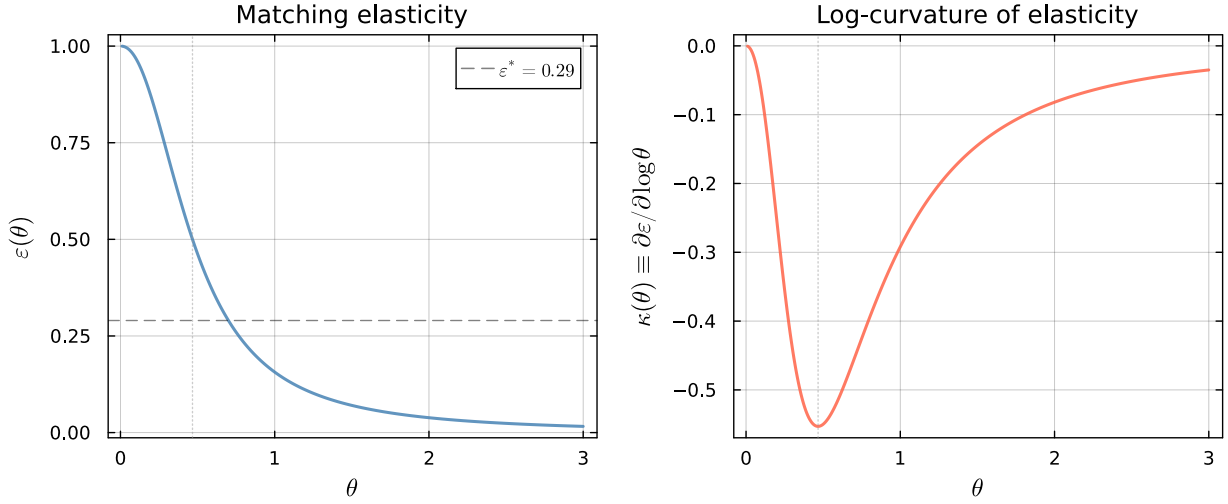


Figure 3: Matching elasticity $\varepsilon(\theta)$ (left) and log-curvature $\kappa(\theta) = \partial\varepsilon/\partial\log\theta$ (right) for the SMCD at $p = p_{\min} \approx 2.21$. Dashed horizontal line: $\varepsilon^* = 0.29$. Vertical dotted line: crossover $\theta = 1/a \approx 0.47$.

4.1. Search Efficiency and the Hosios Condition

Under constant returns to scale, the unemployment elasticity of the matching function is $\eta_u(\theta) \equiv \partial \ln m / \partial \ln u = 1 - \varepsilon(\theta)$. In the standard DMP model with purely flow vacancy costs, Hosios (1990) shows that the decentralized equilibrium is constrained efficient if and only if the worker's bargaining power equals this elasticity, $\alpha_L = \eta_u(\theta) = 1 - \varepsilon(\theta)$. When

⁶The log approximation $\log f(\theta) \approx \log f^* + \varepsilon^* \Delta - \frac{1}{2} p \varepsilon^* (1 - \varepsilon^*) \Delta^2$, with $\Delta = \log(\theta/\theta^*)$, shows that the second-order departure from Cobb-Douglas is proportional to p . Setting $p = p_{\min}$ minimizes this departure while satisfying the probability bounds.

$\alpha_L > \eta_u$, workers extract more surplus than their matching contribution warrants, wages are too high, and equilibrium unemployment exceeds the efficient level.

Under the SMCD, η_u is not a constant. Combining (2) with $\eta_u(\theta) \equiv 1 - \varepsilon(\theta)$ gives

$$\eta_u(\theta) = \frac{(a\theta)^p}{1 + (a\theta)^p},$$

which is strictly increasing in θ , ranging from zero as $\theta \rightarrow 0$ to one as $\theta \rightarrow \infty$. The Hosios condition is therefore state-dependent: calibrating $\alpha_L = \eta_u(\theta^*)$ at the steady state, the economy has $\alpha_L > \eta_u(\theta)$ in recessions (low θ), so workers' bargaining power exceeds their matching contribution in states where unemployment is already elevated. This countercyclical wedge is a structural consequence of bounded probabilities. Section 6.2 generalizes the condition to account for fixed matching costs and quantifies the resulting unemployment gap.

Cobb-Douglas suppresses this mechanism entirely: $\eta_u^{CD} = 1 - \varepsilon_{CD}$ is constant, so Hosios holds uniformly or fails uniformly regardless of the cycle. HRW qualitatively replicates the countercyclical pattern but mismeasures its level. At the pre-break sample mean $\bar{\theta} = 0.67$, HRW implies $\eta_u^{HRW} \approx 0.39$ while SMCD implies $\eta_u \approx 0.60$ — a gap of 0.21 in the efficient bargaining power. Any welfare assessment built on HRW's η_u is misspecified by construction.

5. Empirical Application

5.1. Structural break in the log-linear matching relationship

The preliminary Cobb-Douglas estimate in Section 3 pools seventy-four years of data that span large institutional changes in how jobs are posted and filled. We test whether the log-linear relationship between f_t and θ_t is stable over the full sample using a QLR sup-Wald test for an unknown structural break. Whereas a standard Wald test requires a pre-specified break date, the QLR procedure computes a Wald statistic at every candidate date in a trimmed interior of the sample and takes the supremum. Because maximizing over

candidate dates inflates the test statistic relative to a single-date Wald test, inference relies on the nonstandard asymptotic distribution of [Andrews \(1993\)](#), which we implement via a moving-block bootstrap.

Procedure. Let $y_t = \log f_t$ and $x_t = \log \theta_t$. We fit $y_t = \alpha + \beta x_t + u_t$ on the full sample and test for a break in (α, β) at an unknown date τ . For each candidate τ in a trimmed grid (15% from each end, yielding 629 candidates), we estimate the regression separately on $[1, \tau]$ and $[\tau + 1, n]$, compute Newey–West HAC covariance matrices with bandwidth $L = 7$, and form the Wald statistic $W(\tau) = \Delta(\tau)'[V_1(\tau) + V_2(\tau)]^{-1}\Delta(\tau)$ where $\Delta(\tau) = b_1(\tau) - b_2(\tau)$. The QLR statistic $\sup_{\tau} W(\tau)$ has a nonstandard asymptotic distribution under an unknown break date; we obtain p-values from a moving block bootstrap (MBB, block length 14, $B = 499$ draws) applied to centered residuals from the null model.

Results. Figure [A.3](#), reported in the appendix, plots $W(\tau)$ over time. The sup-Wald statistic is $\sup W = 745.97$ with a bootstrap p-value of 0.000. The estimated break date is October 2008, coinciding precisely with the Lehman Brothers collapse and the onset of the global financial crisis.

Table [2](#) reports the OLS estimates on each subsample. The intercept shifts from -0.691 to -1.144 , a change of -0.453 in logs. At any given tightness, the job-finding rate post-October 2008 is approximately $e^{-0.453} \approx 0.64$ times its pre-2008 level — a persistent 36% decline in matching efficiency. The tightness elasticity $\hat{\epsilon}$ falls modestly from 0.318 to 0.283, indicating that hiring became slightly less responsive to vacancy creation after the crisis.

Table [2](#) reveals that the structural break is overwhelmingly a *level shift*: $\hat{A}$ drops from 0.501 to 0.319 (a 36% decline) while $\hat{\epsilon}$ falls only from 0.318 to 0.283. In the DMP framework, a decline in A shifts the Beveridge curve outward without rotating it, because slope depends on the matching elasticity and separation rate, both essentially unchanged. This is consistent with [Lubik \(2013\)](#), who show the post-2008 outward movement reflects a structural decline in match efficiency, and with [Elsby et al. \(2015\)](#), who identify lateral Beveridge-curve shifts as the dominant source of unemployment-vacancy dynamics. The post-GFC matching efficiency

Table 2: Estimated log-linear matching relationship, pre- and post-October 2008.

	Pre-break (Jan 1951 – Sep 2008)	Post-break (Oct 2008 – Dec 2025)
Intercept $\hat{\alpha}$	−0.691	−1.144
Tightness elasticity $\hat{\varepsilon}$	0.318	0.283
Implied level $\hat{A} = e^{\hat{\alpha}}$	0.501	0.319
Unemployment elasticity $\hat{\eta}_u = 1 - \hat{\varepsilon}$	0.682	0.717
CD lower bound $\hat{A}^{1/\hat{\eta}_u}$	0.36	0.20

Note: Newey–West HAC, $L = 7$. Break identified by QLR sup-Wald with MBB p-values.

decline has been attributed to compositional shifts and mismatch (Barnichon and Figura, 2015; Şahin et al., 2014). Lange and Papageorgiou (2020) show that the decline in hires was driven almost entirely by collapsing matching efficiency rather than by changes in search intensity.

5.2. Horse race: SMCD vs. Cobb-Douglas vs. HRW

We now estimate all three matching functions separately on the pre-break (January 1951 – September 2008) and post-break (October 2008 – December 2025) subsamples and compare fit. Full-sample estimates, which pool the two regimes and are reported for completeness in Figure A.1 in the appendix, show qualitatively similar rankings.

All three models are estimated by nonlinear least squares (NLS), minimizing $\sum_{t=1}^T [f_t - \hat{f}(\theta_t; \boldsymbol{\psi})]^2$ in levels of the job-finding rate. Estimating all models against the same objective ensures a fair comparison of fit. SMCD has three free parameters $\boldsymbol{\psi} = (s, a, p)$, estimated subject to the probability-bound constraints $s \leq 1$ and $sa \leq 1$. HRW has one unconstrained parameter (ν). Cobb-Douglas has two parameters (A, α) and is estimated by NLS in levels without any probability constraint; the log-OLS solution — which minimizes the sum of squared log residuals — serves as the starting point for the NLS routine and is not itself reported as a final estimate. Fit is measured in levels of f by RMSE, MAE, AIC, and BIC; AIC and BIC penalize the additional parameters of SMCD relative to CD and HRW. For

Cobb-Douglas, the NLS and log-OLS estimates are close: the largest deviation is $\Delta\hat{\varepsilon} = 0.019$ in the pre-break subsample, with negligible differences post-break, and the implied $q > 1$ threshold $\hat{\theta}_{\min}$ is unchanged.⁷

Table 3: Matching function fit statistics, pre- and post-October 2008.

Period	Model	Params	RMSE	MAE	AIC	BIC	$q>1$ (%)
Pre-2008	SMCD	3	0.0564	0.0448	-3978.8	-3965.2	0.0
	CD	2	0.0442	0.0349	-4318.0	-4308.9	12.0
	HRW	1	0.0674	0.0549	-3735.9	-3731.4	0.0
Post-2008	SMCD	3	0.0313	0.0238	-1400.1	-1390.2	0.0
	CD	2	0.0284	0.0224	-1441.5	-1434.8	9.4
	HRW	1	0.0573	0.0482	-1158.8	-1155.4	0.0

Note: $n = 693$ pre-break, $n = 203$ post-break. $q>1$ % is the fraction of months where the estimated model implies $q > 1$; zero by construction for SMCD and HRW.

Figure 4 plots the fitted job-finding rate against the data for each subsample. The post-break scatter lies uniformly below the pre-break scatter at any given θ , consistent with the 36% decline in matching efficiency identified in Table 2.

⁷Both f_t and θ_t are constructed from labor market stocks, so they share measurement error in u_t and are simultaneously determined. Two considerations limit the impact on the comparison. First, the NLS estimator is consistent when θ_t is predetermined with respect to the matching-function shock: unemployment is a slow-moving state variable whose current value reflects past separations and job-finding, not the contemporaneous matching realization. Second, as Lange and Papageorgiou (2020) show, unobserved time-varying matching efficiency biases parametric estimators upward, but this bias affects all three specifications equally and therefore does not alter the relative rankings we report. The exercise is a calibration in the tradition of Petrosky-Nadeau and Zhang (2017), not a causal identification; the DMP exercise of Section 6 provides complementary discipline by estimating structural parameters on the globally solved ergodic distribution.

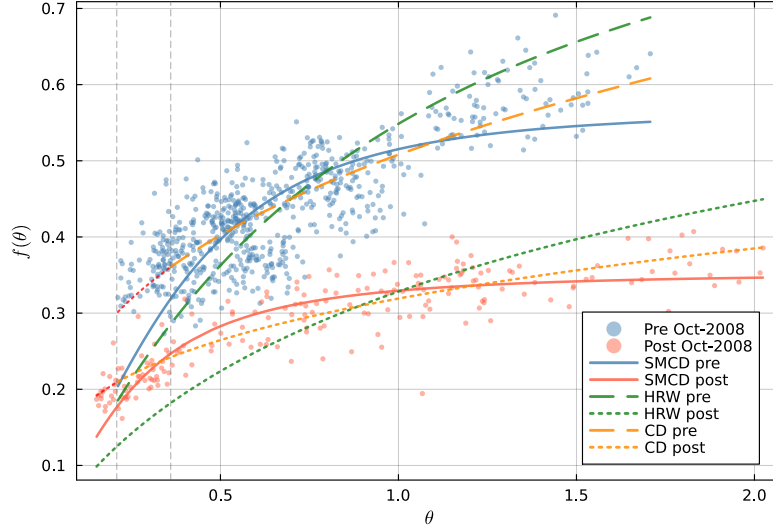


Figure 4: Job-finding rate $f(\theta)$: SMCD (solid), CD (dashed), and HRW (dotted), pre-break (blue) and post-break (red). Grey scatter: data. Red dotted segment: CD in the region $\theta < \theta_{\min}$ where fitted $q > 1$. Vertical gray lines: $\theta_{\min}$.

Table 3 and Figure 4 deliver three findings. First, CD achieves the lowest RMSE and best information criteria in both subsamples. CD is estimated without probability constraints, so it can fit the data by extrapolating into the region where $q > 1$; the horse race rewards this extrapolation with lower residuals in recession months where CD’s fitted vacancy-filling rate exceeds one. SMCD pays a modest cost for guaranteeing validity everywhere: RMSE is 27% higher pre-break and 10% higher post-break. The gap narrows after 2008 because the post-break data span a wider range of tightness (from COVID troughs near $\theta = 0.1$ to the post-pandemic boom above $\theta = 2$), and SMCD handles this dispersion better than a pure power law.

Second, the $q > 1$ column makes CD’s advantage concrete. In the pre-break subsample, CD predicts a vacancy-filling probability exceeding one in 12.0% of months; post-break, 9.4%. In those months, CD achieves a lower residual for f while simultaneously predicting an impossible q . SMCD never fits better than CD in terms of RMSE—even on the restricted sample where both are valid probability models, CD retains a small edge. The correct reading is that SMCD fits *nearly as well* as CD while remaining valid everywhere: a modest RMSE

cost for strict bound compliance. The natural alternative a practitioner might consider—censoring CD by capping q at one—introduces a C^0 kink at the censoring threshold that precludes the smooth global solution of the Bellman equation on which the quantitative exercise of Section 6 relies. Typically, when practitioners need a bounded matching function, they apply HRW (Hagedorn and Manovskii, 2008; Petrosky-Nadeau and Zhang, 2017, 2021). The SMCD proposed here respects the same bounds while offering the calibration flexibility that HRW lacks.

Third, HRW fits substantially worse than both CD and SMCD (RMSE ratios of 1.19 and 1.83). Despite satisfying the probability bounds, HRW’s single-parameter structure prevents it from jointly matching the level and elasticity of job-finding, forcing it to compromise on both. Bound satisfaction alone is insufficient; the additional calibration flexibility of SMCD over HRW translates directly into a large improvement in fit.

Table 4: Estimated parameters, pre- and post-October 2008.

Period	SMCD					CD		HRW	
	s	a	p	sa	$\bar{\varepsilon}$	A	ε	ν	$\bar{\varepsilon}$
Pre-2008	0.567	1.762	2.386	1.000	0.465	0.508	0.336	1.154	0.631
Post-2008	0.354	2.829	1.879	1.000	0.310	0.319	0.271	0.622	0.567

Note: All models estimated by NLS in levels. $\bar{\varepsilon}$ = sample mean of $\varepsilon(\theta_t; \hat{\psi})$ over subsample observations (constant for CD).

Table 4 shows how each model absorbs the structural break. For Cobb-Douglas, $\hat{A}$ declines from 0.508 to 0.319 (37%) while $\hat{\varepsilon}$ shifts only from 0.336 to 0.271, confirming the level-shift interpretation from Section 5.1. The SMCD parameters tell the same story: $\hat{s}$ falls 38% (from 0.567 to 0.354), mirroring the decline in $\hat{A}$, while $\hat{a}$ rises and the crossover tightness $1/\hat{a}$ shifts leftward, compressing the vacancy-limited regime post-crisis. The binding constraint $\hat{sa} = 1.000$ in both periods means NLS allocates the maximum possible vacancy-

filling capacity.

The $\bar{\varepsilon}$ column exposes HRW’s fundamental limitation most sharply. Because $\hat{\nu}$ is the sole parameter, it must simultaneously control the level of $f(\theta)$ and its slope. Pre-2008, the NLS solution lands at $\hat{\nu} = 1.154$, implying a sample-average elasticity of 0.631 — roughly double CD’s estimate of 0.336. Post-2008, $\hat{\nu}$ collapses to 0.622, and the implied $\bar{\varepsilon} = 0.567$ remains far above CD’s 0.271. HRW’s ν is not measuring the matching elasticity; it is absorbing the efficiency level at the cost of a severely distorted curvature profile, which explains the large fit gap relative to both SMCD and CD. The SMCD average elasticities (0.465 pre, 0.310 post) exceed CD’s constant $\hat{\varepsilon}$ as well, but for a different reason: SMCD’s elasticity profile is genuinely nonlinear, rising toward one at low tightness and falling toward zero at high tightness, so recession observations where θ is small pull the arithmetic average above the value that log-linear estimation would assign to the typical observation.

5.3. The Two-Plateau Soft-Min Cobb-Douglas Matching Function

The single- p SMCD dominates HRW conditional on respecting probability bounds, but Cobb-Douglas retains a fit advantage in Table 3 that originates almost entirely from its extrapolation into the $q > 1$ region. A natural one-parameter extension of the SMCD eliminates this residual advantage by assigning independent curvature to each matching regime. Define the *two-plateau soft-min Cobb-Douglas* (2P-SMCD) matching function by

$$f(\theta) = f_c \cdot \frac{\theta}{\theta_c} \left[\frac{2}{1 + (\theta/\theta_c)^p} \right]^{1/p}, \quad p = \begin{cases} p_L & \theta \leq \theta_c \\ p_R & \theta > \theta_c, \end{cases} \quad (4)$$

with parameters $(f_c, \theta_c, p_L, p_R)$, where $f_c > 0$, $\theta_c > 0$, and $p_L, p_R > 0$. The parameter θ_c is the crossover tightness and $f_c = f(\theta_c)$ is the job-finding rate there. The single- p SMCD is recovered at $p_L = p_R$, with $\theta_c = 1/a$ and $f_c = s/2^{1/p}$. The extension adds exactly one genuinely new parameter: the asymmetry $p_L \neq p_R$. The crossover θ_c was already present in the single- p model as $1/a$.

The two tail caps are independently controlled. As $\theta \rightarrow \infty$, $f(\theta) \rightarrow s_R \equiv f_c \cdot 2^{1/p_R}$; as $\theta \rightarrow 0$, $q(\theta) \equiv f(\theta)/\theta \rightarrow q_{\max} \equiv (f_c/\theta_c) \cdot 2^{1/p_L}$. Setting p_L large makes the vacancy-filling probability rise sharply toward $q_{\max}$ in slack markets without imposing any restriction on how quickly the job-finding rate saturates in tight markets, which is governed by p_R alone. This decoupling has a direct economic reading. The parameter p_L governs how quickly vacancy-filling saturates as workers grow scarce, while p_R governs how quickly job-finding saturates as vacancies grow scarce. The crossover θ_c is the balance point at which the two sides of the market contribute equally to matching ($\varepsilon(\theta_c) = 1/2$): to its left, in slack markets, matches respond more to vacancies; to its right, in tight markets, they respond more to searchers. There is no *a priori* reason the pace at which each margin congests should be symmetric across these regimes. The two-plateau relaxes exactly this restriction, at the cost of a single parameter.⁸

For either branch, $\varepsilon(\theta) = 1/(1 + (\theta/\theta_c)^p)$. Direct computation gives

$$\frac{\varepsilon}{1 - \varepsilon} = \frac{1/(1 + (\theta/\theta_c)^p)}{(\theta/\theta_c)^p/(1 + (\theta/\theta_c)^p)} = \left(\frac{\theta_c}{\theta}\right)^p,$$

and therefore

$$\text{logit } \varepsilon(\theta) \equiv \log \frac{\varepsilon}{1 - \varepsilon} = -p \log \left(\frac{\theta}{\theta_c}\right).$$

The logit of the matching elasticity is linear in $\log \theta$, with slope $-p$ and intercept zero at θ_c (where $\varepsilon = 1/2$). For the single- p SMCD, a single line with slope $-p$ describes the entire elasticity profile. For the 2P-SMCD, the profile is piecewise linear: slope $-p_L$ for $\theta < \theta_c$ and slope $-p_R$ for $\theta > \theta_c$, joined at $(\log \theta_c, 0)$. The two slopes are separately identified by

⁸A globally smooth curvature profile could accommodate the same asymmetry, but only by adding several parameters and forfeiting both the analytic probability bounds of Proposition 2 and the closed-form inverse q^{-1} that the free-entry step of the DMP exercise relies on (Appendix C). The two-plateau is the most parsimonious departure from constant curvature—a two-segment linear spline in $(\log \theta, \text{logit } \varepsilon)$ space—and, because it strictly nests the single- p form at $p_L = p_R$, the asymmetry is a testable restriction rather than free tuning.

data on each side of the crossover. Notably, θ_c plays a dual role: it sets the logit-intercepts ($p_{L,R} \log \theta_c$) and marks the kink where curvature shifts between regimes (Figure 5).

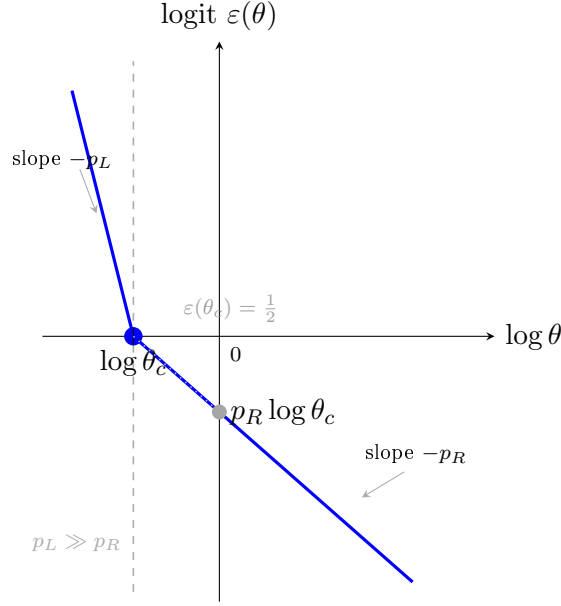


Figure 5: Two-plateau SMCD in $(\log \theta, \text{logit } \varepsilon)$ space, calibrated to pre-2008 estimates $\theta_c = 0.417$ and $p_R = 0.88$; $p_L = 4$ is used for legibility (pre-2008 estimate $p_L = 26.2$ is near-vertical). The dotted line extends the right segment to the y -axis, marking the intercept $p_R \log \theta_c$.

This characterization also clarifies the relationship to the CES discussed in Section 4: the 2P-SMCD produces a piecewise-linear profile that no single- p specification can replicate. The 2P-SMCD is thus a genuinely new functional form, outside the CES class entirely. Moreover, the generalization from the single- p form is intuitive: the single- p SMCD forces one curvature parameter to govern both the vacancy-filling cap in slack markets and the job-finding cap in tight markets, while the 2P-SMCD assigns each regime its own curvature. Proposition 2 confirms that this additional flexibility preserves all four desiderata.

Proposition 2 (Properties of the 2P-SMCD). *The two-plateau matching function defined in (4) satisfies the following four properties.*

- (i) Constant returns to scale. $m(\lambda u, \lambda v) = \lambda m(u, v)$ for all $\lambda > 0$.
- (ii) Continuously differentiable and concave. $f(\theta)$ is C^1 on $(0, \infty)$, smooth on each branch, and globally concave.

- (iii) Joint calibration. For any feasible triple $(f^*, \varepsilon^*, \theta^*)$ with $f^* \leq \min\{1, \theta^*\}$ and $\varepsilon^* \neq 1/2$, there exist parameters (f_c, θ_c, p_R) — or (f_c, θ_c, p_L) — that achieve $f(\theta^*) = f^*$ and $\varepsilon(\theta^*) = \varepsilon^*$.
- (iv) Bounded probabilities. $f(\theta) \leq s_R$ and $q(\theta) \leq q_{\max}$ for all $\theta > 0$. Setting $s_R \leq 1$ and $q_{\max} \leq 1$ ensures $f, q \in (0, 1)$.

Proof. (i) $\theta = v/u$ is homogeneous of degree zero, so $m(\lambda u, \lambda v) = \lambda u \cdot f(\lambda v/\lambda u) = \lambda m(u, v)$. (ii) On each open branch, $f''(\theta) < 0$ by the same calculation as Lemma 2: with $z = \theta/\theta_c$, $h''(z) = -(p+1)z^{p-1}(1+z^p)^{-2-1/p} < 0$ for $p = p_L$ on the left and $p = p_R$ on the right. At θ_c , both branches yield $f(\theta_c) = f_c$. Differentiating and evaluating at $z = 1$ gives $f'(\theta_c) = (f_c/\theta_c) \cdot 2^{1/p} \cdot 2^{-1/p-1} = f_c/(2\theta_c)$, which is independent of p ; hence $f'(\theta_c^-) = f'(\theta_c^+)$ and f is C^1 . A C^1 function that is strictly concave on each of two abutting intervals is globally concave (the secant inequality decomposes at the junction into two within-branch inequalities). (iii) For $\varepsilon^* < 1/2$ (calibration point on the right branch, $\theta^* > \theta_c$): the elasticity condition gives $(\theta^*/\theta_c)^{p_R} = (1 - \varepsilon^*)/\varepsilon^*$, which pins p_R given θ_c ; the level condition then gives $f_c = f^*/(2(1 - \varepsilon^*))^{1/p_R}$. The case $\varepsilon^* > 1/2$ is symmetric on the left branch. (iv) On the right branch, $f(\theta) = f_c(\theta/\theta_c)[2/(1 + (\theta/\theta_c)^{p_R})]^{1/p_R} \leq f_c \cdot 2^{1/p_R} = s_R$. The left branch and the vacancy-filling bound are analogous. $\square$

Local Cobb-Douglas behavior follows directly from (ii) and (iii). Since $\varepsilon(\theta)$ is continuous everywhere, including at θ_c , then at any calibration point θ^* where $\varepsilon(\theta^*) = \varepsilon^*$, this yields $d \log m = (1 - \varepsilon^*)d \log u + \varepsilon^* d \log v$ to first order. Thus, the 2P-SMCD is locally Cobb-Douglas at θ^* . The second-order deviation is $-\frac{1}{2}p_{\text{branch}} \varepsilon^*(1 - \varepsilon^*)\Delta^2$, where p_{branch} is the curvature of the branch containing θ^* .

In the single- p SMCD, a single parameter p must satisfy both bound constraints simultaneously; at the US calibration the vacancy-filling bound forced $p_{\min} \approx 2.21$, producing curvature $|\kappa(\theta^*)| \approx 0.46$ even though the calibration point lies in the worker-limited regime. The 2P-SMCD decouples the two constraints: the right cap $s_R \leq 1$ is governed by p_R alone, while p_L independently controls the left cap $q_{\max} \leq 1$. The minimum feasible p_R — the

smallest value for which a finite p_L satisfying $q_{\max} \leq 1$ exists — is approximately 0.97, reducing curvature at θ^* to $|\kappa| \approx 0.20$, roughly half the single- p value. The corresponding right-branch ceiling is $s_R \approx 0.569$, a 22% improvement over the single- p cap of $s \approx 0.467$. Figure 6 plots all four specifications at the common calibration. In the empirical horse race of Section 5.4, the NLS estimates are unconstrained by the sample-mean calibration, so the estimated s_R is free to exceed this theoretical ceiling.

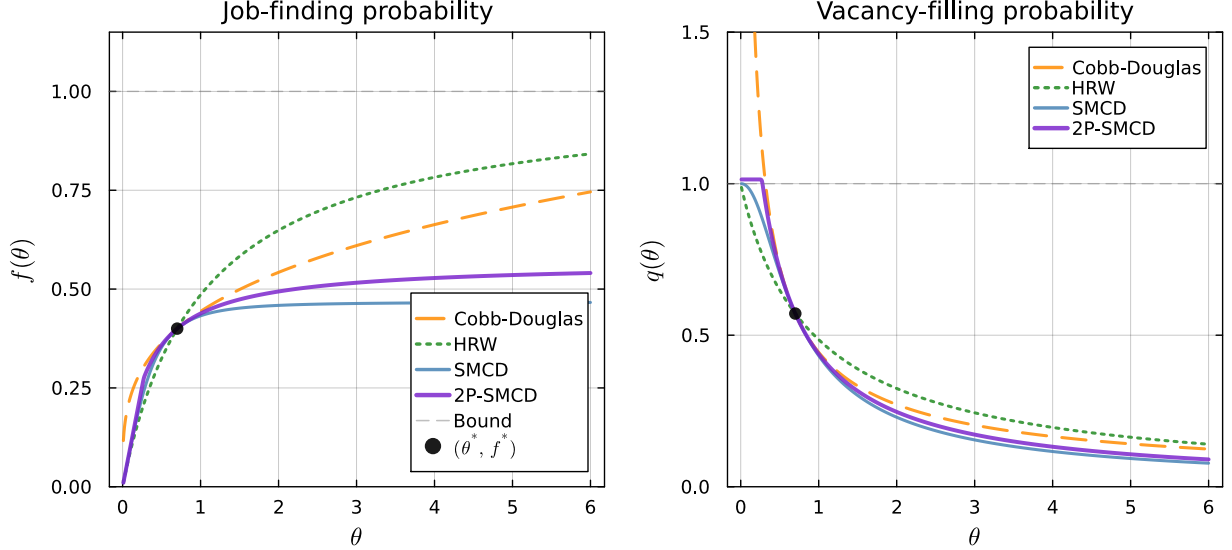


Figure 6: Job-finding probability $f(\theta)$ (left) and vacancy-filling probability $q(\theta)$ (right) for CD (dashed), HRW (dotted), single- p SMCD (solid, light), and 2P-SMCD (solid, dark). All calibrated to $(\theta^*, f^*, \varepsilon^*) = (0.70, 0.40, 0.29)$ where feasible; HRW matches f^* at θ^* only. Filled circle: calibration point.

5.4. Horse Race: Adding the Two-Plateau SMCD

Before comparing fit statistics it is useful to define the domain on which CD constitutes a valid probability model. The estimated function $f(\theta) = A\theta^\varepsilon$ satisfies $q(\theta) = A\theta^{\varepsilon-1} \leq 1$ only for $\theta \geq \theta_{\min} \equiv A^{1/(1-\varepsilon)}$, and $f(\theta) \leq 1$ only for $\theta \leq \theta_{\max} \equiv A^{-1/\varepsilon}$. Pre-break, $(\theta_{\min}, \theta_{\max}) = (0.36, 7.50)$; post-break, $(0.21, 68.1)$. The upper bound $\theta_{\max}$ is never reached in either subsample, so the binding constraint is always $\theta_{\min}$: CD violates $q \leq 1$ in 12.0% of pre-break months and 9.4% of post-break months. The lower bound shifts left after the Lehman collapse because the decline in matching efficiency A widens the range of tightness for which CD overestimates the vacancy-filling rate.

The 2P-SMCD is estimated by nonlinear least squares over $(\theta_c, f_c, p_L, p_R)$, subject to the bounds $s_R \equiv f_c \cdot 2^{1/p_R} \leq 1$ and $q_{\max} \equiv (f_c/\theta_c) \cdot 2^{1/p_L} \leq 1$; the $q > 1$ violation rate is therefore zero by construction. Table 5 reports standard fit statistics alongside a restricted RMSE*, computed exclusively on months with $\theta > \theta_{\min}$ — the region where CD also predicts valid probabilities and the comparison is on equal terms. Table 6 reports the estimated parameters and derived caps.

Table 5: Fit statistics for the 2P-SMCD and Cobb-Douglas, pre- and post-October 2008.

Period	Model	Par	RMSE	RMSE*	MAE	$q > 1$ (%)
Pre-2008	CD	2	0.0442	0.0447	0.0349	12.0
	2P-SMCD	4	0.0550	0.0465	0.0433	0.0
Post-2008	CD	2	0.0284	0.0293	0.0224	9.4
	2P-SMCD	4	0.0344	0.0298	0.0260	0.0

Note: $n = 693$ pre-break, $n = 203$ post-break. RMSE* computed on months with $\theta > \theta_{\min}$ ($\theta_{\min} = 0.36$ pre-break, 0.21 post-break), where CD also satisfies $q \leq 1$. CD RMSE* exceeds its full-sample RMSE because the excluded low- θ months are recession observations where CD's unconstrained extrapolation fits well.

Table 6: Estimated 2P-SMCD parameters, derived tail caps, and average tightness elasticity, pre- and post-October 2008.

Period	θ_c	f_c	p_L	p_R	s_R	$q_{\max}$	$\bar{\varepsilon}$
Pre-2008	0.417	0.361	26.18	0.882	0.791	0.888	0.493
Post-2008	0.325	0.231	10.04	1.439	0.373	0.761	0.381

Note: $s_R = f_c \cdot 2^{1/p_R}$; $q_{\max} = (f_c/\theta_c) \cdot 2^{1/p_L}$. $\bar{\varepsilon} = T^{-1} \sum_t \varepsilon(\theta_t; \hat{\psi})$: average tightness elasticity over the subsample, from the piecewise analytical formula (Section 5.3).

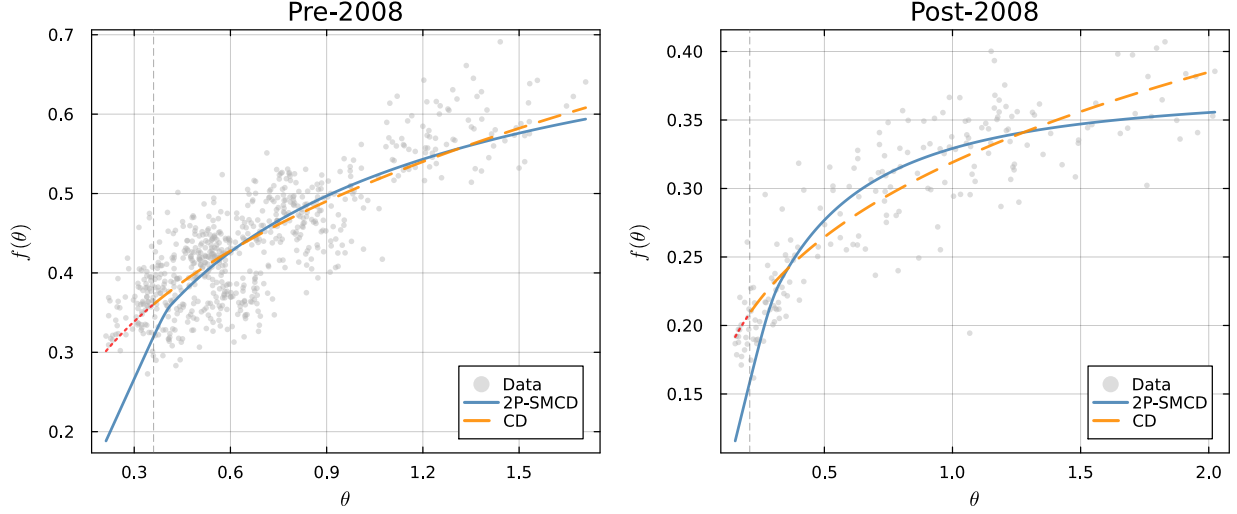


Figure 7: Job-finding rate $f(\theta)$: 2P-SMCD (solid blue) and CD (dashed orange), pre-break (left) and post-break (right). Grey scatter: data. Red dotted segment: CD in the region $\theta < \theta_{\min}$ where fitted $q > 1$. Vertical gray line: $\theta_{\min}$.

Tables 5–6 and Figure 7 deliver three findings.

First, CD achieves lower full-sample RMSE in both subsamples, by the same margin as the single- p SMCD. Adding a fourth parameter does not overturn this ranking when the full time series — including recession troughs where CD extrapolates into $q > 1$ — is used as the criterion.

Second, as in the single- p horse race (Section 5.2), virtually the entire RMSE advantage of CD is concentrated in months where it is not a valid probability model. Pre-break, the full RMSE gap (2P-SMCD minus CD) is $0.0550 - 0.0442 = 0.0108$; the restricted gap narrows to $0.0465 - 0.0447 = 0.0018$, recovering 83% of the advantage. Post-break, the analogous figures are a full gap of $0.0344 - 0.0284 = 0.0060$ and a restricted gap of $0.0298 - 0.0293 = 0.0005$, so 92% is accounted for by the $q > 1$ months. Within the admissible domain $\theta > \theta_{\min}$, the 2P-SMCD fits within 4.0% of CD’s RMSE pre-break and 1.7% post-break.

Third, the estimated parameters reveal strong asymmetry: $\hat{p}_L$ is large (26.2 pre-break, 10.0 post-break) while $\hat{p}_R < 2$ (0.88 and 1.44). The vacancy-filling probability rises sharply to its cap on the left branch, while the job-finding rate saturates gradually on the right. A single- p specification cannot simultaneously produce a sharp left-side bound and a smooth

right-side saturation. The concentrated NLS objective (Appendix D) confirms that θ_c , f_c , and p_R are sharply identified, while p_L is identified from below—the data reject $p_L < 13.2$ (pre-break) and $p_L < 4.7$ (post-break) at 95%—but flat from above: once the left branch is steep enough for near-Leontief behavior, additional steepness does not improve fit. The estimated $\hat{s}_R = 0.791$ pre-break exceeds the theoretical minimum-curvature ceiling of 0.569 from Section 5.3, confirming that the NLS solution exploits the full tightness distribution.

The structural break carries a direct message for applied users. Researchers should use the sub-period estimates in Tables 5 and 6 corresponding to their application’s temporal focus; full-sample estimates are defensible only if the structural model endogenously explains the efficiency decline through search intensity, worker composition, or mismatch (Şahin et al., 2014; Barnichon and Figura, 2015). The 2P-SMCD ensures matching probabilities remain bounded regardless of which parameter vector is used.

6. A Quantitative DMP Exercise

The horse race establishes that the 2P-SMCD fits matching data as well as CD while respecting probability bounds, and strictly dominates HRW. Here we ask how much the distinction between the two bounded specifications matters for efficiency and unemployment crises. We embed the pre-break 2P-SMCD estimates in a DMP economy with Pissarides (2009) fixed matching costs and compare its business-cycle behavior to an otherwise identical HRW economy. Both are calibrated by simulated method of moments to the *same* three targets—the relative volatility of unemployment, mean tightness, and the wage elasticity—so any remaining difference is attributable solely to the *shape* of the matching function.

Three results emerge. First, bounded matching probabilities imply a modified Hosios condition (Proposition 3) under which the efficient bargaining weight $\alpha_L^H(\theta)$ depends on tightness and on the fixed-cost share ϕ , collapsing toward zero as $\phi \rightarrow 1$. This is a new analytical result: under the standard (flow-cost-only) Hosios condition, the efficient weight is the constant $\eta_u(\theta)$; with the two-part cost of Pissarides (2009), it becomes $\alpha_L^H(\theta) =$

$(1 - \phi(\theta))\eta_u(\theta)/[1 - \phi(\theta)\eta_u(\theta)]$. A planner allocation reveals a sign-reversing unemployment gap: the 2P-SMCD economy is inefficiently tight on average but generates 12.9 pp of excess unemployment in crises, while HRW sits near constrained efficiency. Second, the near-Leontief recession branch generates vacancy shutdowns and fat-tailed crisis episodes—the [Petrosky-Nadeau and Zhang \(2017\)](#) channel, which persists under a matching function *jointly calibrated* to level and elasticity targets. Third, the Hosios wedge is roughly twice as volatile under the 2P-SMCD as under HRW, because its steeper elasticity gradient amplifies the gap between the constant bargaining weight and the state-dependent efficient share.

This exercise demands a global solution, for a reason deeper than nonlinearity alone. The job-finding rate is bounded and concave, so unemployment rises faster in recessions than it recedes in expansions. A first-order approximation is symmetric by construction, so it misses this asymmetry. Higher-order perturbation does not repair it, because the equilibrium tightness policy is not analytic. The vacancy non-negativity constraint makes firms stop posting once a filled match is worth less than the hiring cost. This puts a kink in the tightness policy at the shutdown threshold. The two-plateau crossover θ_c leaves the policy only once-differentiable at a second point. Both points lie inside the ergodic set the economy visits. A Taylor expansion around the deterministic steady state therefore has a radius of convergence bounded by the distance to these singularities. Adding orders need not improve accuracy in the recessions that produce our results, and [Petrosky-Nadeau and Zhang \(2017\)](#) find that it can worsen it. We therefore solve the model globally, by time iteration on the job-creation condition. The continuation value is evaluated by Gauss–Hermite quadrature over the continuous productivity distribution rather than a discrete Markov chain. Following [Petrosky-Nadeau and Zhang \(2021\)](#), we estimate the structural parameters on the globally simulated ergodic distribution, so the same nonlinear dynamics that generate the results also discipline the calibration.

6.1. Environment and equilibrium

Time is discrete and a period is one month. A filled job produces output z_t , where $\log z_t$ follows an AR(1) process.⁹ Unemployed workers and vacancies meet according to the matching function $m(u, v)$, generating a job-finding rate $f(\theta) = m(1, \theta)$ and a vacancy-filling rate $q(\theta) = m(1, \theta)/\theta$ at tightness $\theta = v/u$; matches dissolve at the exogenous monthly rate δ .

Firms post vacancies subject to the two-part hiring cost of [Pissarides \(2009\)](#): a flow cost k for each period a vacancy is open, plus a fixed cost κ paid once a match forms.¹⁰ The expected cost of a hire is $k/q(\theta) + \kappa$, and we summarize its composition by the fixed-cost share

$$\phi(\theta) = \frac{\kappa}{\kappa + k/q(\theta)}. \quad (5)$$

Free entry drives the value of an open vacancy to zero whenever vacancies are posted, $q(\theta_t)(J_t - \kappa) = k$: a firm pays the flow cost k to hold the vacancy open and, upon meeting a worker at rate $q(\theta_t)$, pays the fixed cost κ and obtains a filled job worth J_t . The fixed cost is incurred immediately upon matching, so it enters undiscounted. This timing feature accords with [Pissarides \(2009\)](#) and [Christiano et al. \(2016\)](#), whose free-entry condition is likewise $q(J - \kappa)$ equals the posting cost.¹¹ The value of a filled job satisfies the Bellman equation

$$J_t = \beta \mathbb{E}_t[z_{t+1} - w_{t+1} + (1 - \delta) J_{t+1}], \quad (6)$$

⁹We estimate the productivity process on the pre-break sample: quarterly HP-filtered log output per worker has autocorrelation 0.757 and innovation standard deviation 0.0084, which map to monthly $\rho_z = 0.911$ and $\sigma_z = 0.0053$. The conditional expectation is evaluated by Gauss–Hermite quadrature over the continuous innovation distribution; see Section 6.3 and [Appendix C](#).

¹⁰Here κ denotes the fixed matching cost, distinct from the log-curvature $\kappa(\theta)$ of Section 4.

¹¹Our monthly timing separates matching from production: a match formed in period t first produces in $t + 1$, so J_t in (6) is a forward, discounted value (the β multiplies the entire continuation). [Christiano et al. \(2016\)](#), at quarterly frequency, let matching and production coincide within the period. In both cases, the cost κ paid undiscounted at the match.

where β is the monthly discount factor. The value of an open vacancy is $V_t^v = -k + q(\theta_t)(J_t - \kappa)$. Free entry drives it to at most zero, $V_t^v \leq 0$, or equivalently

$$J_t \leq \kappa + \frac{k}{q(\theta_t)}, \quad \theta_t \geq 0, \quad \theta_t \left(\kappa + \frac{k}{q(\theta_t)} - J_t \right) = 0. \quad (7)$$

Whenever tightness is positive, free entry holds with equality, $J_t = \kappa + k/q(\theta_t)$, and combining it with (6) yields the job-creation condition

$$\kappa + \frac{k}{q(\theta_t)} = \beta \mathbb{E}_t \left[z_{t+1} - w_{t+1} + (1 - \delta) \left(\kappa + \frac{k}{q(\theta_{t+1})} \right) \right], \quad (8)$$

which equates the hiring cost of a filled job (left-hand side) to its present discounted value (right-hand side). Because $q(\theta) \leq q_{\max} < \infty$, the cost floor $\kappa + k/q_{\max}$ cannot be undercut: once J_t falls below it no firm posts and tightness hits the corner $\theta_t = 0$ —the second and third parts of (7), a treatment of the vacancy non-negativity constraint in the manner of [Petrosky-Nadeau and Zhang \(2017\)](#).

Wages are set by Nash bargaining with worker weight α_L ,

$$w_t = (1 - \alpha_L) b + \alpha_L (z_t + k \theta_t), \quad (9)$$

with b the flow value of nonemployment. Unemployment evolves as

$$u_{t+1} = u_t + \delta(1 - u_t) - f(\theta_t) u_t. \quad (10)$$

Given the exogenous productivity process $\{z_t\}$, an equilibrium is a set of sequences $\{\theta_t, u_t, w_t\}$, with $\theta_t \geq 0$, satisfying the free-entry complementarity (7), the wage equation (9), and the law of motion (10) in every period. Because productivity is the only state, tightness and the wage are time-invariant functions of it, $\theta_t = \theta(z_t)$ and $w_t = w(z_t)$; we solve for these policy functions globally (Section 6.3).

This shutdown corner is absent from the unbounded Cobb-Douglas benchmark, where $q(\theta) \rightarrow \infty$ as $\theta \rightarrow 0$ and the cost floor collapses to κ ; under a bounded q —whether 2P-SMCD or HRW—it is a live event, where a capped filling rate leaves a deep recession no way to fill jobs faster. Yet the fat right tail of unemployment in Section 6.4 does not hinge on the corner itself: in the near-Leontief regime around it, the flat q makes tightness and job finding fall convexly with productivity, and the low- f months pile up in the unemployment stock. The shutdown is the limiting case of this contemporaneous nonlinearity, not a separate anticipatory force.

6.2. Efficiency: the Hosios condition with fixed matching costs

Section 4.1 established that bounded probabilities make the Hosios condition $\alpha_L = \eta_u(\theta)$ state-dependent: it can hold at most one point of the cycle, and the resulting wedge is countercyclical. We now generalize this result to account for the fixed matching cost κ of the present environment, which modifies the efficient bargaining weight. The quantitative analysis then shows that this produces a sign-reversing unemployment gap.

Proposition 3. *The decentralized equilibrium is constrained efficient if and only if the worker’s bargaining weight satisfies*

$$\alpha_L = \alpha_L^H(\theta) \equiv \frac{(1 - \phi(\theta)) \eta_u(\theta)}{1 - \phi(\theta) \eta_u(\theta)}, \quad (11)$$

with $\phi(\theta)$ given by (5). At $\phi = 0$ this reduces to the standard condition $\alpha_L = \eta_u(\theta) = 1 - \varepsilon(\theta)$ of Section 4.1.

The intuition is that only the flow component $k/q(\theta)$ of the hiring cost is congestion-sensitive: it is the margin through which an extra vacancy lengthens the queue and lowers q for other firms. The elasticity of the total hiring cost $k/q(\theta) + \kappa$ with respect to tightness is $(1 - \phi(\theta)) \eta_u(\theta)$ —the flow-cost congestion elasticity η_u diluted by the flow share $1 - \phi$. The fixed matching cost κ is paid per match irrespective of market congestion and therefore carries

no externality; as the cost mix tilts toward it ($\phi \rightarrow 1$), the congestion externality vanishes and the efficient worker share falls to zero. The derivation of the dynamic social planner’s problem with the two-part cost is in [Appendix B.2](#). The standard condition $\alpha_L = \eta_u(\theta)$ obtains in the flow-cost-only environment of [Pissarides \(2000, ch. 8\)](#). As best as we know, (11) has not previously been stated. [Mangin and Julien \(2021\)](#) generalize the Hosios condition along a different margin—nonlinear surplus splitting—while retaining a single flow posting cost; their modification and ours are complementary and could in principle be combined.

6.3. Calibration and solution

We adopt the estimates for the matching-function parameters from [Section 5.4](#) at the pre-break period: the 2P-SMCD parameters $(\theta_c, f_c, p_L, p_R)$ and the HRW parameter ν (with $s = a = 1$). This cleanly separates identification: the shape of the matching technology is disciplined by the matching data, while the three structural parameters (b, α_L, k) —which the matching data cannot speak to—are estimated inside the model. The separation rate is $\delta = 0.035$ ([Shimer, 2005](#)), the monthly discount factor is $\beta = e^{-0.04/12}$. We set the fixed-cost share to $\phi = 0.90$ at the steady state, pinning κ from k and $q(\theta^*)$ via (5). A share this high is motivated by [Christiano et al. \(2016\)](#), who estimate a search-theoretic DSGE model by Bayesian impulse response matching and find the fixed component accounts for 94% of the cost of meeting a worker; our 90% is a conservative lower bound of their estimate.

We use three moments from the simulated data to identify the parameter triple (b, α_L, k) : the relative volatility of unemployment $\sigma(u)/\sigma(z) = 10.8$, the mean tightness $\bar{\theta} = 0.67$, and the wage elasticity $\varepsilon_{w,y} = 0.52$. The flow value b governs the size of the match surplus and hence the amplification of unemployment to productivity through the mechanism of [Hagedorn and Manovskii \(2008\)](#); the flow cost k sets the level of tightness—and thereby the average operating point on the matching function; and the worker’s bargaining weight α_L controls how strongly wages track productivity. Volatility is measured as HP-filtered *proportional* deviations from the mean, following [Petrosky-Nadeau and Zhang \(2017\)](#), rather than log deviations. Although unemployment is bounded away from zero in the data and in

the model’s ergodic distribution—so the log transform is well-defined for u —the proportional-deviation convention preserves the cardinal interpretation of rates and maintains a single consistent metric across all variables, including vacancies and tightness, which do reach zero at the shutdown corner.¹² By targeting $\bar{\theta}$ and $\sigma(u)/\sigma(z)$ rather than mean unemployment and $\sigma(\theta)/\sigma(z)$, we let both the mean unemployment rate and the tightness volatility ratio be *outcomes* that can differ across specifications, revealing how the matching function’s shape affects both the level and the higher moments of unemployment.

Targeting $\bar{\theta}$ rather than $\bar{u}$ implies that the model will overshoot the data’s mean unemployment. Given the horse-race estimates, $f(0.67) \approx 0.44$ – 0.45 . Flow balance pins steady-state unemployment at $u^* = \delta/(\delta + f(\bar{\theta})) \approx 7.3\%$ —already above the data’s 5.6%. The ergodic mean lies weakly above even this steady-state value: because $f(\theta)$ is concave, Jensen’s inequality implies $\mathbb{E}[f(\theta)] < f(\mathbb{E}[\theta])$, so workers find jobs less efficiently on average than they would at the average level of tightness, pushing $\bar{u}$ above u^* —consistent with the simulated 7.4–7.6%. The gap between the model and empirical mean of unemployment reflects the two-state model’s coarseness: all transitions flow through a single separation rate $\delta = 0.035$ and a single job-finding rate f , abstracting from job-to-job moves and participation-margin flows that lower observed unemployment. Any DMP model with these primitives produces $\bar{u} \approx 7$ – 7.5% . If we instead targeted $\bar{u} = 5.6\%$, the model would need $\bar{f} \approx 0.59$, above the average rate implied by the horse-race estimates. The overshoot is the price of letting the matching data discipline the mean rather than the unemployment rate.

The amplification target $\sigma(u)/\sigma(z) \approx 10.8$ follows the convention of [Petrosky-Nadeau and Zhang \(2017\)](#) for quantifying the unemployment-volatility puzzle of [Shimer \(2005\)](#). In a

¹²The two transforms agree to first order— $\log(x_t/\bar{x}) = (x_t - \bar{x})/\bar{x} - \frac{1}{2}[(x_t - \bar{x})/\bar{x}]^2 + \dots$ —so their standard deviations coincide when fluctuations are small. They diverge in the tails: the log stretches the left tail ($\log(x/\bar{x}) \rightarrow -\infty$ as $x \rightarrow 0$) and compresses the right (sublinear growth). For tightness, which collapses toward zero in crises, log deviations inflate $\sigma(\log \theta)$; for unemployment, whose right tail fattens, they compress the spikes and deflate $\sigma(\log u)$. The net effect biases $\sigma(\log \theta)/\sigma(\log u)$ upward. Proportional deviations are the linear transform: they impose no curvature, so the measured asymmetry is the model’s, not the metric’s. They also preserve the cardinal interpretation of rates—a rise in u from 5% to 10% is a 100% deviation, not $\log 2 \approx 0.69$ —and are well-defined at the vacancy-shutdown corner where $\theta = 0$.

single-shock model the only question is which unconditional volatility ratio the shock should reproduce; matching a conditional elasticity—attributing to the model only the productivity-driven share of fluctuations—would deliver a far tamer economy in which the global nonlinearities that separate the 2P-SMCD from HRW would scarcely operate. We reach the amplification regime not through a near-productivity value of nonemployment, as in Hagedorn and Manovskii (2008), but through the fixed matching cost of Pissarides (2009): the estimated flow value of nonemployment, $b \approx 0.77$ for the 2P-SMCD (Table 7), lies well below the Hagedorn–Manovskii value of 0.955—which Hall and Milgrom (2008) show implies a Frisch labor-supply elasticity outside any empirical range—and near the 0.71 that Hall and Milgrom obtain by imputing the value of non-work directly.¹³ The amplification is the fixed matching cost’s doing, not a thin surplus wrung from a near-productivity outside option.

Solution algorithm. The model is solved by time iteration on the job-creation condition using Gauss–Hermite quadrature over the continuous log-normal innovation distribution, eliminating the discretization staircase that a finite Markov chain introduces in the tails. Given a guess for the expected continuation value $E(z) = \mathbb{E}[J(z') \mid z]$, each iteration inverts the free-entry condition (8) for $\theta(z)$. If the implied value falls short of $\kappa + k/\bar{q}$ the vacancy-shutdown corner binds and $\theta(z) = 0$.

The three parameters are estimated by simulated method of moments (SMM) on the global solution.¹⁴ We solve for the bargaining weight α_L apart from the two cost parameters, because each target has a parameter it speaks to most directly: b governs the amplification of unemployment, through the size of the match surplus; k governs the level of tightness, through the level of vacancy posting; and α_L governs the cyclicalities of wages, through the share of the procyclical surplus that workers capture. We put this structure to work by

¹³These figures are flow values of nonemployment relative to labor productivity ($\bar{z} = 1$), the object comparable across studies; they are *not* replacement ratios. The implied b/w is higher, about 0.82, because the low estimated bargaining weight ($\alpha_L \approx 0.10$) leaves the wage only slightly above the outside option.

¹⁴The three-moment system is exactly identified, so the estimator matches the targets exactly; following the quantitative DMP literature (Hagedorn and Manovskii, 2008; Petrosky-Nadeau and Zhang, 2017) we treat the three data moments as calibration targets and do not conduct inference.

holding α_L fixed, solving for the cost pair (b, k) that reproduces $\sigma(u)/\sigma(z)$ and $\bar{\theta}$, and then turning the single dial α_L —a higher weight makes wages more procyclical—until the simulated wage elasticity matches the data, re-solving (b, k) at each turn. [Appendix C](#) details the solver and the estimation loop. [Table 7](#) reports the estimated parameters and their fit to the three targets.

Table 7: DMP calibration: pre-break targets and calibrated parameters.

	2P-SMCD	HRW	Target
Flow value of nonemployment b	0.769	0.698	—
Worker bargaining weight α_L	0.102	0.061	—
Flow vacancy cost k	0.323	0.454	—
Fixed matching cost κ	4.36	6.23	—
Fixed-cost share ϕ	0.90	0.90	0.90
$\sigma(u)/\sigma(z)$	10.64	10.85	10.77
Mean tightness $\bar{\theta}$	0.671	0.670	0.671
$\varepsilon_{w,y}$	0.52	0.52	0.52

Note: Monthly model. $\delta = 0.035$, $\beta = e^{-0.04/12}$. Moments in HP-filtered proportional deviations from the mean. All three moments targeted; $\sigma(\theta)/\sigma(z)$ and $\bar{u}$ are untargeted outcomes ([Table 8](#)).

6.4. Results

[Table 8](#) reports business-cycle moments for the two calibrated economies alongside the pre-break data. By construction both match $\sigma(u)/\sigma(z)$, $\bar{\theta}$, and the wage elasticity; the differences lie in untargeted objects. Mean unemployment, now an outcome, is 7.4% under the 2P-SMCD and 7.6% under HRW—both above the data’s 5.6%, for the reasons discussed in [Section 6.3](#). More revealing is the tightness volatility ratio: $\sigma(\theta)/\sigma(z) = 14.7$ under the 2P-SMCD against 17.1 under HRW, both below the data value of 21.4. That the 2P-SMCD requires *less* tightness volatility to deliver the same unemployment volatility reflects its nonlinear left branch: in the near-Leontief regime, a given decline in θ maps into a disproportionately large rise in unemployment, so the same $\sigma(u)/\sigma(z)$ is achieved with less volatile θ . The job-finding rate is correspondingly less volatile under the 2P-SMCD

($\sigma(f)/\sigma(z) = 7.5$ versus 11.0), because its bounded upper plateau saturates f in booms while HRW’s smoother profile keeps the matching elasticity relatively high throughout. The unemployment–productivity correlation is weaker under the 2P-SMCD (-0.52 versus -0.80): its state-dependent, asymmetric u dynamics are poorly summarized by a linear comovement, echoing the weaker correlations [Petrosky-Nadeau and Zhang \(2017\)](#) obtain from a global solution relative to log linearization and closer to the mild relationship in the data.

Two caveats apply to the comparison with data. First, both economies are driven by a single productivity shock, so model variables comove nearly perfectly with z , whereas the data, presumably driven by many shocks, show weak correlations with productivity. This also explains why $\sigma(w)/\sigma(z) = 0.52$ falls below the empirical 0.93 despite matching the wage elasticity $\varepsilon_{w,y} = 0.52$, since $\rho(w, z) \approx 1$ makes relative volatility and elasticity coincide in the model. Second, as the single shock must account for all unemployment variation, the calibrated b is higher than micro estimates would suggest; a multi-shock model would spread the amplification burden across sources.

The unemployment distribution. The sharpest difference is in the shape of the unemployment distribution. Figure 8 plots the simulated density of the monthly unemployment rate under the two calibrations. Panel (a) shows that the 2P-SMCD core is *more concentrated* than HRW’s: its unconditional standard deviation is 0.95 percentage points against 1.06, and its mode is taller and narrower. HRW carries visibly more mass in the range 9–12%. A reader who stopped at panel (a) would conclude that HRW generates more unemployment risk. Panel (b), plotted on a log scale, reverses that impression. Beyond about 12% the 2P-SMCD density dominates, and by the 15% crisis threshold of [Petrosky-Nadeau and Zhang \(2021\)](#) the 2P-SMCD density is roughly ten times higher (the crisis *frequency* is 0.22% versus 0.07%, a factor of 3.3). The skewness of the unemployment rate in levels—13.8 for the 2P-SMCD against 1.8 for HRW (Figure 8)—summarizes this reversal: the 2P-SMCD distribution is tight in the core but carries a long, heavy right tail.

The mechanism is the near-Leontief left branch. In normal times ($\theta > \theta_c$), the bounded

Table 8: Business-cycle moments: data and calibrated models.

		Data	2P-SMCD	HRW
Tightness θ	$\sigma(x)/\sigma(z)$	21.36	14.69	17.09
	$\rho(x, z)$	0.30	0.99	1.00
	autocorr	0.88	0.69	0.70
Unemployment u	$\sigma(x)/\sigma(z)$	10.77	10.64	10.85
	$\rho(x, z)$	-0.29	-0.52	-0.80
	autocorr	0.88	0.72	0.78
	mean (level)	5.62%	7.39%	7.59%
Vacancies v	$\sigma(x)/\sigma(z)$	11.44	11.24	10.60
	$\rho(x, z)$	0.40	0.89	0.88
	autocorr	0.91	0.48	0.42
Job finding f	$\sigma(x)/\sigma(z)$	6.57	7.52	11.02
	$\rho(x, z)$	0.28	0.85	0.99
	autocorr	0.79	0.60	0.70
Wage w	$\sigma(x)/\sigma(z)$	0.93	0.52	0.52
	$\rho(x, z)$	0.62	0.99	1.00
	autocorr	0.80	0.69	0.70

Note: HP-filtered proportional deviations from the mean ($\lambda = 1600$). Data: pre-break subsample, reference variable output per worker; z is model productivity. Model: 200,000-month simulations. Mean unemployment (level) is an untargeted outcome.

vacancy-filling rate saturates and hiring operates near capacity, compressing the core distribution. But when a sufficiently adverse shock pushes θ below θ_c , the flat q leaves no room for faster hiring, and unemployment spills upward along a near-deterministic path: during vacancy shutdown ($\theta = 0$), the law of motion $u_{t+1} = u_t + \delta(1 - u_t)$ drives unemployment up by roughly $\delta \approx 3.5$ percentage points per month. This produces the secondary peaks visible in panel (b) at approximately 3-percentage-point intervals—the “stepping stones” of one-, two-, and three-month shutdown spells. HRW’s smoother matching technology, by contrast, spreads volatility evenly across the cycle: moderate shocks produce moderate unemployment fluctuations throughout, generating broader everyday variation but no catastrophic collapses. Its density decays monotonically with no secondary structure. This is the crisis

mechanism of Petrosky-Nadeau and Zhang (2017) and Petrosky-Nadeau and Zhang (2021), arising from the geometry of the matching function rather than from an extreme surplus calibration.

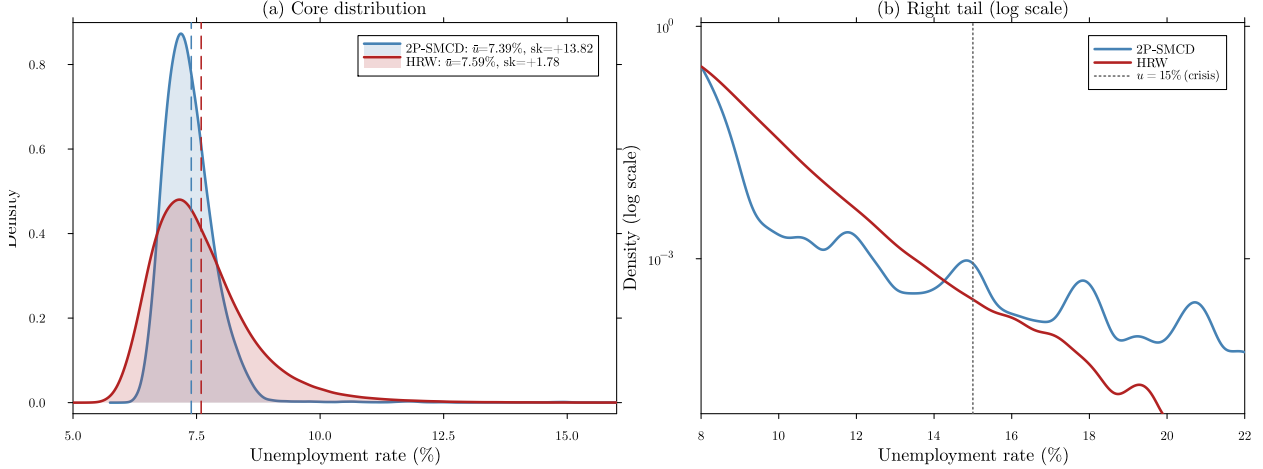


Figure 8: Simulated density of the monthly unemployment rate: 2P-SMCD and HRW. Panel (a): linear scale; dashed lines mark the means. Panel (b): log scale; dotted line marks the 15% crisis threshold.

The Hosios wedge. At the calibration, fixed costs dominate hiring ($\phi = 0.90$), so the efficient worker share of Proposition 3 is low—but the two matching functions deliver very different levels. The simulated mean $\alpha_L^H(\theta)$ is 0.133 under the 2P-SMCD and 0.060 under HRW, close to but not identical to the calibrated weights $\alpha_L = 0.102$ and 0.061. On average the decentralized SMCD economy modestly *under-rewards* workers (mean modified wedge $\alpha_L - \alpha_L^H = -0.031$) while HRW sits essentially at the modified Hosios boundary (wedge $+0.001$), because its elasticity $\eta_u(\theta)$ varies too little over the cycle to open a sizeable wedge. This near-efficiency verdict requires accounting for cost composition: measured against the conventional flow-cost rule $\alpha_L = \eta_u(\theta)$, whose simulated mean is ≈ 0.60 (2P-SMCD) and 0.38 (HRW), the same calibrated $\alpha_L \approx 0.1$ appears to grossly *under-reward* workers—a mean wedge near -0.5 —implying wildly excessive vacancy creation. The fixed-cost correction of Proposition 3, recognizing that the 90%-dominant matching cost carries no congestion externality, lowers the efficient share to ≈ 0.1 and overturns that verdict.

The two specifications differ sharply in the *cyclical* of the wedge. The efficient share

$\alpha_L^H(\theta)$ moves over the cycle while the bargained α_L is fixed; the standard deviation of $\alpha_L - \alpha_L^H(\theta)$ is 0.031 under the 2P-SMCD against 0.018 under HRW. The 2P-SMCD compounds the state-dependence of $\eta_u(\theta)$ with that of $\phi(\theta)$, roughly doubling the cyclical variation. Because $\eta_u(\theta)$ falls toward zero as tightness collapses, the efficient share drops below the bargained α_L in deep recessions. The fraction of time in which the wedge flips sign—workers *over*-appropriate—is 14.9% under the 2P-SMCD against 54.2% under HRW, but HRW’s crossings are economically inconsequential (it hovers near zero wedge), while the SMCD’s sign flips are concentrated in deep recessions, where over-appropriation coincides with already-high unemployment.

The unemployment gap. The Hosios wedge in parameter space (α_L versus α_L^H) does not by itself quantify the degree of excess or insufficient unemployment. To answer that question, we solve the planner’s allocation. This consists of swapping α_L in the model with the (state-dependent) $\alpha_L^H(\theta)$ of Proposition 3, fixing the parameters at the calibrated values, and solving the resulting planner allocation as a full dynamic global solution using the same time-iteration solver, Gauss–Hermite quadrature, and shock realizations as the decentralized economy, differing only in that the constant bargained weight α_L is replaced state by state with the efficient weight $\alpha_L^H(\theta)$. The per-period gap $u_t^{\text{dec}} - u_t^{\text{pln}}$ then isolates the welfare cost of constant- α_L bargaining.

Under the 2P-SMCD, the unemployment gap flips sign. On average, the decentralized economy under-bargains ($\alpha_L = 0.102 < \alpha_L^H \approx 0.133$), so firms over-hire and unemployment is *too low*: mean $\bar{u}^{\text{dec}} = 7.4\%$ against $\bar{u}^{\text{pln}} = 7.7\%$, a gap of -0.30 percentage points. But in crisis episodes ($u \geq 15\%$), the mean gap flips to $+12.9$ percentage points of *excess* unemployment—larger than the 9.4-pp peak gap in the representative (median-peak) episode below, because the distribution of crisis peaks is right-skewed and a few extreme spells pull the mean above the median peak. In the near-Leontief left regime, $\varepsilon(\theta) \rightarrow 1$ and $\eta_u(\theta) \rightarrow 0$, so the modified Hosios condition prescribes $\alpha_L^H \rightarrow 0$ —the congestion externality vanishes and the planner would slash the shadow “wage” to approximately b , hiring aggressively. The

decentralized economy, locked into $\alpha_L = 0.102$, cannot adjust. The planner never shuts down vacancy posting; the decentralized economy does so in six of twenty-five productivity states.

Figure 9 isolates a representative crisis episode—the spell whose peak unemployment is closest to the median peak across all 127 crisis spells in the simulation. Before the crisis, the decentralized economy runs slightly below the planner’s unemployment (blue shading in panel a): the calibrated $\alpha_L = 0.102$ lies below the efficient share $\alpha_L^H \approx 0.133$, so firms capture too much of the surplus and over-hire. When a sequence of adverse productivity shocks pushes tightness into the near-Leontief left regime, the decentralized economy shuts down vacancy posting entirely— $\theta = 0$ for three months (panel b, pink band)—while the planner, whose $\alpha_L^H(\theta) \rightarrow 0$ slashes the effective wage share, keeps tightness near 0.4 and vacancies open throughout. The result is a 9.4-percentage-point excess unemployment gap at the peak: decentralized unemployment spikes to 18.0% against the planner’s 8.6%. After shutdown ends, the pattern reverses: decentralized tightness overshoots the planner as the over-hiring regime reasserts itself, and unemployment drops back below the efficient level within about ten months.

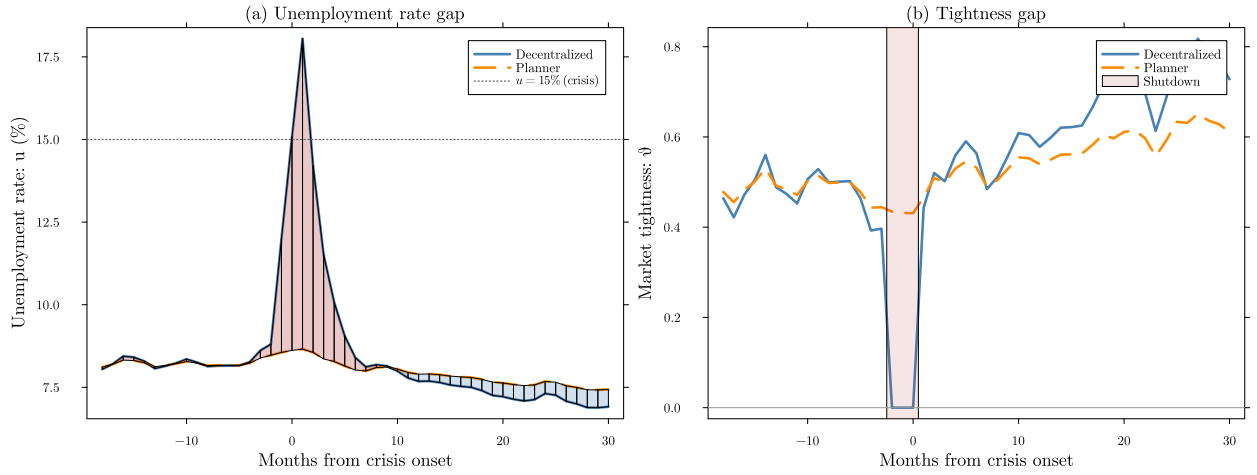


Figure 9: Representative crisis episode under the 2P-SMCD: decentralized (solid) versus planner (dashed) allocation. Panel (a): unemployment rate; shading marks the gap (red: excess, blue: insufficient); dotted line: 15% crisis threshold. Panel (b): market tightness; pink band: vacancy-shutdown region ($\theta = 0$).

The HRW economy tells a very different story. Its calibrated $\alpha_L = 0.061$ sits essentially at $\alpha_L^H = 0.060$, so mean planner and decentralized unemployment nearly coincide (7.4% versus

7.6%, gap +0.19 pp). The gap splits roughly evenly between excess (55%) and insufficient (45%) unemployment, with no clear direction. In crisis episodes the gap is +6.2 pp—less than half the SMCD value—because HRW’s smooth logistic elasticity profile never produces the sharp $\alpha_L^H \rightarrow 0$ collapse. The gap is more linearly countercyclical ($\rho(u_{\text{gap}}, z) = -0.79$ versus -0.37 for SMCD), confirming that HRW’s inefficiency is spread thinly across the cycle rather than concentrated in the states where it hurts most.

Recalibration equalizes the moments the two economies are asked to match but cannot equalize objects that depend on curvature: tail risk, Hosios-wedge volatility, and crisis-episode unemployment gaps all depend on p_L and θ_c , which are estimated from matching data, not chosen to fit business-cycle moments. Because p_L is identified only from below (Appendix D), Appendix D.1 recomputes these objects across the identified range $p_L \in [13, 40]$: the crisis gap (12.1–13.4 pp) and unemployment skewness (13.3–14.0) are stable throughout, so the headline results turn on the asymmetry $p_L \gg p_R$ rather than on the exact left-branch curvature.

This inefficiency relates to, but is distinct from, the model-free sufficient-statistics approach of [Michaillat and Saez \(2021, 2024\)](#). [Michaillat and Saez \(2021\)](#) derive a general formula for the *Beveridgean unemployment gap* $u - u^*$, where the efficient rate u^* equates the marginal social costs of unemployment and vacancies. Their formula requires three sufficient statistics: the elasticity of the Beveridge curve (the elasticity of v with respect to u along the uv -locus—an equilibrium object folding in separations and job creation, distinct from the matching-function elasticity $\varepsilon(\theta)$ that governs α_L^H here), the social cost of unemployment, and the cost of recruiting. The efficient tightness θ^* under this general formula need not equal one; it depends on the cost ratio and the Beveridge elasticity. Crucially, the formula that maps these sufficient statistics to u^* does not require specifying the wage-setting mechanism: the Beveridge curve, measured directly from data on u and v , already embeds whatever wage protocol generated the equilibrium.

[Michaillat and Saez \(2024\)](#) show that the general formula admits a powerful special

case that can be immediately operationalized with readily available data. Under three assumptions—the Beveridge curve is an approximate rectangular hyperbola ($uv \approx \text{const}$, an empirical regularity), one recruiter is required per vacancy so that a jobseeker and a vacancy are equally socially costly, and jobsearch has no social value—the sufficient-statistics formula collapses to $u^* = \sqrt{uv}$: the market is efficient at tightness $\theta = 1$, inefficiently tight when $\theta > 1$, and inefficiently slack when $\theta < 1$.

In our environment, the fixed matching cost breaks the one-recruiter-per-vacancy symmetry, so the efficient allocation is characterized not by $\theta = 1$ but by the modified Hosios weight $\alpha_L^H(\theta)$ of Proposition 3. Our “inefficiently tight on average” is accordingly a statement about *over-hiring* ($\bar{u}^{\text{dec}} < \bar{u}^{\text{pln}}$ because $\alpha_L < \overline{\alpha_L^H}$), not about θ exceeding one. The novelty is the *cyclical reversal*: the bounded matching function drives $\eta_u(\theta) \rightarrow 0$ in crises, collapsing the efficient share below the fixed bargained weight and turning average over-hiring into large excess unemployment precisely when unemployment is already high.¹⁵

Two caveats frame these welfare statements. First, the bargained weight α_L is a calibrated convention—disciplined to reproduce the wage elasticity—rather than an object identified from matching data. Second, the sign-reversing gap is driven jointly by the shape of the matching function and by the *acyclicity* of the bargained share: a wage protocol delivering a procyclical worker share, such as alternating-offer bargaining, would let α_L track $\alpha_L^H(\theta)$ downward in recessions and attenuate the crisis gap. The welfare verdict is therefore conditional on Nash bargaining with a constant weight. The more robust feature is the direction and state-dependence of the efficient share $\alpha_L^H(\theta)$, which follows from the matching technology alone.

¹⁵A further contrast concerns the efficient tightness itself. In the symmetric-cost framework of [Michaillat and Saez \(2024\)](#) it is the constant $\theta^* = 1$, invariant to productivity. In our model the flow and fixed recruiting costs are fixed in units of output and do not scale with the temporary shock z , so the planner’s efficient tightness $\theta^{\text{pln}}(z)$ is procyclical—a higher z raises match surplus against a fixed recruiting cost. The sufficient-statistics gap thus measures the position of θ relative to a fixed efficient point, whereas our gap measures the decentralized allocation relative to a procyclical $\theta^{\text{pln}}(z)$.

Implementation. The planner allocation can be decentralized by a state-dependent hiring tax or subsidy levied on the firm per filled match. The relevant margin is the firm’s job-creation condition (8): the firm posts vacancies until the hiring cost $\kappa + k/q(\theta)$ equals the expected value of a filled job, which depends on flow profit $z - w$. The subtlety is that Nash bargaining renegotiates the wage in response to the tax. A per-period levy T on the firm reduces the total surplus available for splitting from $z - b + k\theta$ to $z - b + k\theta - T$; Nash bargaining then lowers the negotiated wage by $\alpha_L T$, so the firm absorbs only a fraction $1 - \alpha_L$ of the tax (Appendix B.2.1). The transfer that equates the firm’s effective labor cost $w + (1 - \alpha_L)T$ with the planner wage $w^{\text{pln}} = b + \alpha_L^H(\theta)(z - b + k\theta)$ is

$$T(\theta, z) = \frac{\alpha_L^H(\theta) - \alpha_L}{1 - \alpha_L} \cdot (z - b + k\theta),$$

where the factor $1/(1 - \alpha_L)$ corrects for bargaining incidence. When $T > 0$ (normal times, $\alpha_L^H \approx 0.133 > \alpha_L = 0.102$), the tax discourages over-hiring; when $T < 0$ (crises, $\alpha_L^H \rightarrow 0 < \alpha_L$), the subsidy incentivizes vacancy creation in the states where the decentralized economy shuts down. The sign flip in T mirrors the sign flip in the unemployment gap.

The relationship between matching-function shape and policy design connects to Landais et al. (2018), who characterize optimal unemployment insurance under constant matching elasticity and risk-averse workers in the Merz (1995) large-family framework. In their environment, the efficiency cost of generous benefits is lower in recessions because the congestion externality remains material throughout the cycle, making optimal UI countercyclical. The present model instead features variable matching elasticity, risk-neutral workers, and a hiring tax/subsidy rather than UI as the policy instrument. Under the SMCD’s state-dependent elasticity, the congestion externality vanishes in crises ($\eta_u \rightarrow 0$), so the efficiency-side case for discouraging job creation disappears at the point where unemployment is highest—the planner instead wants a hiring subsidy.

Implementation requires observing θ in real time and knowing the matching function’s

shape, both of which the horse-race estimates provide. The practical nuance is that the tax/subsidy schedule $T(\theta, z)$ is nonlinear and state-dependent, requiring more information than a flat hiring credit or a fixed replacement rate.

The welfare results depend on ϕ , the fixed-cost share, set to $\phi = 0.90$ following [Christiano et al. \(2016\)](#). Because the modified Hosios condition compresses α_L^H toward zero as $\phi \rightarrow 1$, [Appendix D.2](#) recalibrates the entire model at $\phi = 0.80$. The qualitative verdict is unchanged: the crisis-episode gap and unemployment skewness are virtually unchanged, and the low calibrated bargaining weight is confirmed to come from the high fixed-cost share rather than from any intrinsic feature of the matching function.

7. Discussion: The Continuous-Time Objection

One might object that the probability-bound violations motivating the SMCD are an artifact of discrete time and disappear in continuous-time formulations. In a standard continuous-time DMP model, the matching function $m(u, v)$ specifies a *flow rate* of matches per unit time, and the associated job-finding and vacancy-filling objects $f(\theta) = m(1, \theta)$ and $q(\theta) = m(1, \theta)/\theta$ are hazard rates rather than probabilities. Hazard rates are non-negative but not bounded above by one: a Poisson matching process can deliver $f(\theta) > 1$ per unit time without logical contradiction, since the implied per-period probability $1 - e^{-f(\theta)\Delta t}$ is in $(0, 1)$ for any finite $\Delta t > 0$. On this view, the CD function $f(\theta) = A\theta^{1-\alpha}$ is a valid hazard-rate specification for all θ , and the violations documented in [Table 1](#) are an artifact of applying a continuous-time object directly as a monthly probability.

The objection is technically correct but overstates what continuous time resolves, and its surviving force turns out to bear entirely on the vacancy-filling rate q at low tightness: even under the exact hazard-to-probability mapping the job-finding rate f is bounded, but q is not. The first difficulty is empirical. Matching functions are estimated by confronting model-implied objects with data at monthly frequency — the standard practice from [Shimer \(2005\)](#) to [Hagedorn and Manovskii \(2008\)](#). A model that predicts $f(\theta) > 1$ at realized tightness

values is misspecified at the calibration frequency, regardless of whether it is written in continuous or discrete time.

The second difficulty cuts against the objection on its own terms. The correct link between a continuous-time hazard $\lambda_f(\theta)$ and the monthly probability is $f_t = 1 - e^{-\lambda_f(\theta)}$, not the first-order approximation $f_t \approx \lambda_f(\theta)$. Applied to the CD hazard $\lambda_f(\theta) = A\theta^{1-\alpha}$, this yields

$$f_t(\theta) = 1 - e^{-A\theta^{1-\alpha}}, \quad (12)$$

which lies in $(0, 1)$ for all θ and has a variable elasticity $\varepsilon(\theta) = A(1-\alpha)\theta^{1-\alpha}/(e^{A\theta^{1-\alpha}} - 1)$ that declines with θ . The practitioner who applies this mapping consistently has abandoned the log-linear matching function in favor of a bounded, variable-elasticity specification sharing the core features of the SMCD. But (12) bounds only f , not q : since $q(\theta) = f(\theta)/\theta \rightarrow \infty$ as $\theta \rightarrow 0$, the exponential specification still predicts $q > 1$ below a threshold that includes the deepest recession observations. At the COVID trough $\theta \approx 0.15$, (12) with $\hat{A} = 0.45$ and $\hat{\alpha} = 0.71$ implies $q \approx 1.53$. The SMCD bounds both $f \leq s$ and $q \leq sa$ simultaneously.

Ultimately, the strongest case for bounded matching functions is quantitative, not philosophical. Section 6 shows that the near-Leontief branch of the 2P-SMCD — a direct consequence of the vacancy-filling bound — drives unemployment skewness (13.8 versus 1.8 under HRW), fat-tailed crisis risk, and a Hosios efficiency wedge that flips sign across the cycle. These properties are invisible to constant-elasticity specifications and to first-order solution methods. Whether or not one accepts the probability cap as a theoretical primitive, its quantitative implications for unemployment dynamics and welfare are large enough to justify the additional parameter.

8. Conclusion

The SMCD matching function resolves a tension between the two standard specifications in search theory. Cobb-Douglas fits the data well but predicts impossible vacancy-filling prob-

abilities in recessions. HRW respects the probability bounds but cannot calibrate jointly to a target job-finding rate and elasticity. The SMCD does both, satisfying four desiderata with only three parameters and closed-form calibration formulas: constant returns, smoothness and concavity, exact joint calibration, and bounded matching probabilities.

On U.S. data, a QLR sup-Wald test identifies a structural break at October 2008, a 36% decline in matching efficiency with little change in elasticity. Across both subsamples the single- p SMCD strictly dominates HRW. Cobb-Douglas attains a lower unconditional RMSE, but overwhelmingly from the 12% and 9% of months in which it predicts $q > 1$. Excluding those months removes 83% of its pre-break advantage and 92% of its post-break advantage. The two-plateau extension adds one parameter for regime-specific curvature and closes the rest. The horse race thus pins down the shape of the matching function while respecting the bounds Cobb-Douglas violates.

The general-equilibrium payoff turns on this shape, under a clean division of identification. We fix the shape from the horse race and estimate the remaining structural parameters on separate aggregate moments, so the results reflect the matching technology rather than the calibration. Under the two-part hiring cost of [Pissarides \(2009\)](#), a modified Hosios condition (Proposition 3) measures efficiency. The efficient bargaining weight $\alpha_L^H(\theta) = (1 - \phi)\eta_u / (1 - \phi\eta_u)$ is state-dependent through both the matching elasticity $\eta_u(\theta)$ and the fixed-cost share ϕ . Under Cobb-Douglas this variation is mild. Under the SMCD, whose elasticity collapses in recessions, it is large enough to reverse the sign of the unemployment gap.

Calibrated to the *same* three targets, the two economies diverge sharply. The 2P-SMCD economy is inefficiently tight on average, so firms over-hire, echoing the efficient-unemployment-rate evidence of [Michaillat and Saez \(2024\)](#). In crises the gap reverses to +12.9 pp of excess unemployment, against +6.2 pp under HRW. This sign reversal follows from a constant bargaining weight meeting an efficient share that collapses in recessions. The same near-Leontief branch preserves the vacancy-shutdown channel of [Petrosky-Nadeau](#)

and Zhang (2017) under joint calibration. It makes unemployment far more skewed and crisis-prone, and the planner never shuts down vacancies whereas the decentralized SMCD economy does.

Two extensions are natural. The post-2008 tightness range runs from the COVID trough near $\theta \approx 0.1$ to a post-pandemic boom above $\theta = 2$, spanning both limiting regimes and offering an out-of-sample test. Embedding the SMCD in a model with risk-averse agents would revisit the optimal-UI question of Landais et al. (2018), and confront the vanishing congestion externality of Section 6 with the insurance motive.

Appendix A. Additional Figures

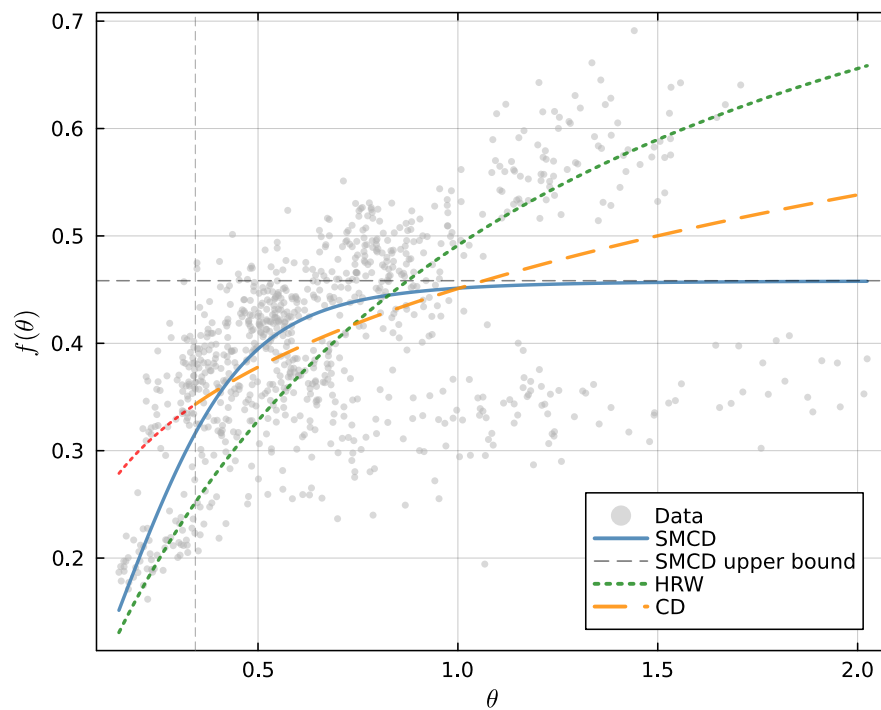


Figure A.1: Full-sample horse race: job-finding rate $f(\theta)$ for SMCD (solid), Cobb-Douglas (dashed), and HRW (dotted), 1951–2025. Grey scatter: data. Dashed horizontal line: SMCD upper bounds.

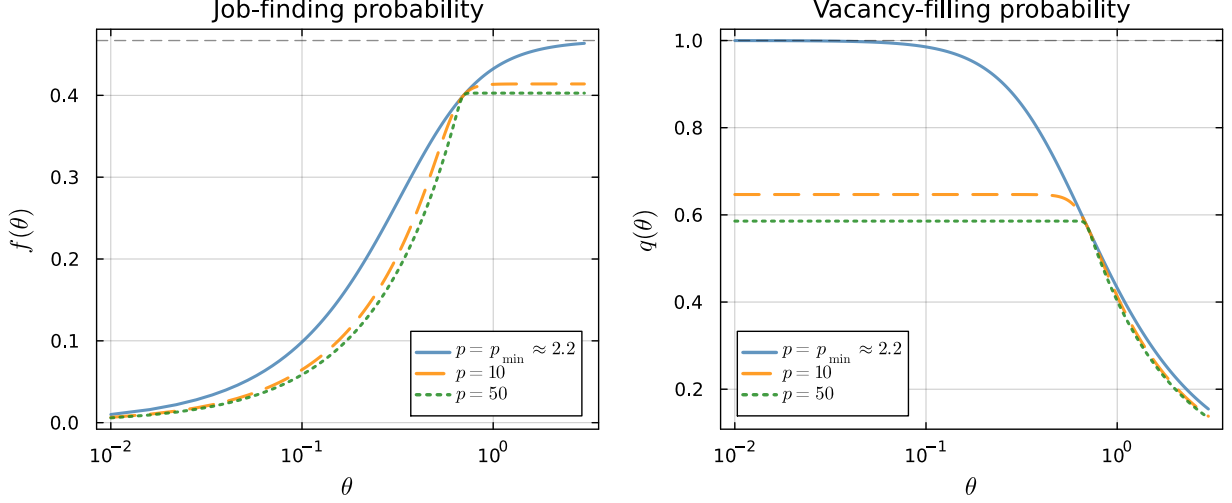


Figure A.2: SMCD job-finding rate (left) and vacancy-filling rate (right) for three values of p . Calibration: $(\theta^*, f^*, \varepsilon^*) = (0.70, 0.40, 0.29)$; (s, a) recalibrated for each p . Dashed lines: asymptotes $f = s$ and $q = sa$ at $p = p_{\min}$.

Appendix B. Proofs and Derivations

Appendix B.1. Concavity Proof

Let $z = a\theta$ and define $h(z) = z(1 + z^p)^{-1/p}$. As $f(\theta) = sh(a\theta)$, $f''(\theta) = sa^2h''(a\theta)$, so it suffices to show $h''(z) \leq 0$ for $z > 0$. Calculations yield

$$h'(z) = (1 + z^p)^{-1/p-1}$$

$$h''(z) = -(p+1)z^{p-1}(1 + z^p)^{-2-1/p} \leq 0 \quad \text{for } z > 0, p > 0.$$

Appendix B.2. The Modified Hosios Condition with Fixed Matching Costs

This appendix derives Proposition 3 from the dynamic planner's problem of the stochastic environment of Section 6.1, following the planner-versus-decentralization logic of Hosios (1990) and Pissarides (2000, ch. 8) extended to the two-part hiring cost. The efficiency condition is a *within-period* arbitrage that holds separately at each productivity state z ; the deterministic steady state ($z_t = \bar{z}$) is the special case, and the continuation values a dynamic derivation carries cancel from the condition. This is why solving the planner's allocation state by state with $\alpha_L^H(\theta)$ —as Section 6.4 does—is exact, not a steady-state approximation.

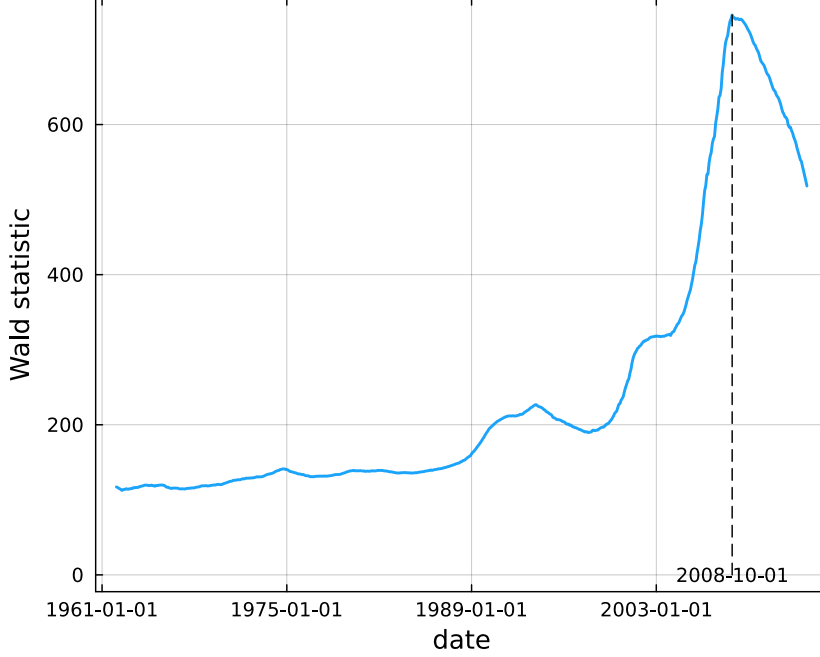


Figure A.3: QLR sup-Wald statistic $W(\tau)$ for a structural break in the log-log matching relationship $\log f_t = \alpha + \beta \log \theta_t + u_t$. Vertical dashed line: estimated break date, October 2008. $\sup W = 745.97$; bootstrap p-value = 0.000.

Decentralization. Let $S(z)$ denote the total surplus of a filled match, $S(z) = J(z) + (W(z) - U)$, the sum of the firm's job value (6) and the worker's net value. Generalized Nash bargaining with worker weight α_L splits it, $W(z) - U = \alpha_L S(z)$ and

$$J(z) = (1 - \alpha_L) S(z). \quad (\text{B.1})$$

The fixed matching cost κ is borne by the firm at the moment of formation and is not part of the split—consistent with the Nash wage (9), in which κ does not appear. Whenever vacancies are posted, free entry drives the value of a vacancy to zero: a vacancy costs k per period and fills at the private rate $q(\theta)$, delivering the match net of the fixed cost, so

$$\frac{k}{q(\theta(z))} = J(z) - \kappa \quad \Longleftrightarrow \quad J(z) = \kappa + \frac{k}{q(\theta(z))}. \quad (\text{B.2})$$

Both (B.1) and (B.2) hold state by state.

Planner. A utilitarian planner chooses tightness to maximize the present value of output net of the flow value of nonemployment and of hiring costs,

$$\max_{\{\theta_t \geq 0\}} \mathbb{E}_0 \sum_{t \geq 0} \beta^t \left[z_t(1 - u_t) + b u_t - k \theta_t u_t - \kappa u_t f(\theta_t) \right], \quad u_{t+1} = u_t + \delta(1 - u_t) - u_t f(\theta_t).$$

Let $\lambda(z)$ be the social value of an additional filled job—the planner’s counterpart of $J(z)$. An extra vacancy raises the flow of matches not at the private rate $q(\theta)$ but at the social rate $\partial m / \partial v = \varepsilon(\theta) q(\theta)$, where $\varepsilon(\theta) = \partial \ln m / \partial \ln v$, so the planner’s vacancy first-order condition equates the flow cost to the social matching rate times the net social value, $k = \varepsilon(\theta) q(\theta) (\lambda(z) - \kappa)$, i.e.

$$\lambda(z) = \kappa + \frac{k}{\varepsilon(\theta(z)) q(\theta(z))}, \quad (\text{B.3})$$

which again holds state by state. Like the private surplus, the social value obeys a forward recursion, $\lambda(z) = (z - b) + \beta(1 - \delta) \mathbb{E}[\lambda(z') \mid z]$, with the *same* discount factor $\beta(1 - \delta)$ and the same flow surplus $z - b$ that drive $S(z)$; the congestion externality enters only through the matching rate $\varepsilon(\theta)q(\theta)$ in (B.3), not through the continuation.

Efficiency. The decentralized and planner allocations coincide at every state when $S(z) = \lambda(z)$: the two surpluses then satisfy an identical recursion, so equality at every z is consistent and both economies choose the same $\theta(z)$. Imposing $S = \lambda$ and combining the within-period relations (B.1), (B.2), and (B.3),

$$(1 - \alpha_L) \left(\kappa + \frac{k}{\varepsilon q} \right) = \kappa + \frac{k}{q}.$$

Writing $x \equiv k/q(\theta)$ and solving for α_L ,

$$\alpha_L = \frac{x(1 - \varepsilon)}{\varepsilon \kappa + x}.$$

Dividing numerator and denominator by $\kappa+x$ and using the fixed-cost share $\phi(\theta) = \kappa/(\kappa+x)$ from (5), together with $\eta_u(\theta) = 1 - \varepsilon(\theta)$, gives

$$\alpha_L^H(\theta) = \frac{(1 - \phi(\theta)) \eta_u(\theta)}{1 - \phi(\theta) \eta_u(\theta)},$$

which is (11). The continuation values have dropped out: efficiency is the within-period requirement that the surplus split neutralize the congestion externality, imposed at each θ separately, so the same $\alpha_L^H(\theta)$ characterizes both the deterministic steady state and every state of the dynamic economy. At $\phi = 0$ (purely flow costs, $\kappa = 0$) it returns the standard condition $\alpha_L = \eta_u(\theta) = 1 - \varepsilon(\theta)$; at $\phi \rightarrow 1$ it tends to zero. Only the flow cost $x = k/q(\theta)$ responds to congestion, so the fixed matching cost dilutes the externality that Nash bargaining must internalize by the factor $1 - \phi(\theta)$.

The vacancy-shutdown corner. Because (B.3) and (B.2) are within-period arbitrage conditions, the non-negativity constraint $\theta \geq 0$ leaves the interior efficiency condition unchanged: it only fixes the states in which posting ceases. The corner binds for the planner and for the market at (generally different) thresholds in $\lambda(z)$ and $J(z)$; wherever both post, (11) characterizes efficiency, and the state-dependent allocation of Section 6.4 imposes $\alpha_L^H(\theta)$ throughout. $\square$

Appendix B.2.1. Decentralizing the planner allocation

This subsection derives the state-dependent hiring tax or subsidy $T(\theta, z)$ that makes the decentralized equilibrium replicate the planner allocation of Proposition 3. The key subtlety is that Nash bargaining renegotiates the wage in response to any tax levied on the firm, so only a fraction of the tax is borne by the firm. The required instrument must compensate for this pass-through.

The Nash wage (9) splits the per-period match surplus $\Sigma \equiv z - b + k\theta$ between worker

and firm:

$$w = b + \alpha_L \Sigma, \quad z - w = (1 - \alpha_L) \Sigma.$$

The fixed matching cost κ is sunk at the point of bargaining and does not appear in Σ : it enters only the free-entry condition (7). Now suppose the government levies a state-dependent per-period transfer $T(\theta, z)$ on the firm ($T > 0$: tax; $T < 0$: subsidy). Nash bargaining splits the reduced surplus $\Sigma - T$, giving

$$w^\tau = b + \alpha_L (\Sigma - T) = w - \alpha_L T.$$

The firm's total per-period labor cost—wage plus tax—is therefore

$$w^\tau + T = w + (1 - \alpha_L) T : \tag{B.4}$$

Nash renegotiation absorbs a fraction α_L of the tax, so only $(1 - \alpha_L)T$ falls on the firm. The planner allocation is replicated when the firm faces the effective wage it would pay under $\alpha_L^H(\theta)$, i.e. $w^\tau + T = b + \alpha_L^H(\theta) \Sigma$. Substituting (B.4) and solving,

$$T(\theta, z) = \frac{\alpha_L^H(\theta) - \alpha_L}{1 - \alpha_L} \cdot (z - b + k\theta).$$

When $\alpha_L^H(\theta) > \alpha_L$ (the decentralized economy over-hires), $T > 0$: the tax discourages vacancy posting. When $\alpha_L^H(\theta) < \alpha_L$ (the decentralized economy under-hires), $T < 0$: the subsidy incentivizes vacancy creation in the states where the congestion externality vanishes.

□

Appendix C. Global Solution and Estimation

This appendix details the two nested numerical steps summarized in Section 6.3: an inner *global* solution of the model by time iteration on the job-creation condition, and an outer estimation that matches moments of the model’s simulated stationary distribution to their empirical counterparts. The two matching functions are solved and estimated identically. Table C.1 collects the numerical settings.

Appendix C.1. Time iteration

Equilibrium tightness solves the job-creation condition (8), a fixed-point problem in the value of a filled job J as a function of the productivity state z . Rather than discretize $\log z$ on a Markov chain, we work with the expected continuation value $E(z) \equiv \mathbb{E}[J(z') \mid z]$ directly, evaluating the conditional expectation by Gauss–Hermite quadrature over the continuous log-normal innovation distribution. This eliminates the discretization staircase in policy functions that a finite grid introduces, which would otherwise distort the tail behavior of the simulated unemployment distribution. The iteration variable $E(z)$ is represented on a 25-point grid in z ; conditional expectations at off-grid productivity realizations during simulation are obtained by linear interpolation. Time iteration works directly with the equilibrium free-entry condition rather than a value function; from an initial guess, each pass performs the following.

1. **Initialize** $E(z)$ at the deterministic steady-state continuation value.
2. **Free-entry inversion.** At each grid point, invert $q(\theta_i) = k/(E_i + \kappa_{\text{net}} - \kappa)$ for tightness, where κ_{net} accounts for the next-period fixed cost; the inverse q^{-1} is analytic for both the two-plateau SMCD and the HRW. Shutdown states are handled by the ramp introduced below.
3. **Flow profit.** Substitute the Nash wage (9) to obtain $z_i - w_i = (1 - \alpha_L)(z_i - b) - \alpha_L k \theta_i$.
4. **Howard improvement.** Take $K = 15$ partial policy-evaluation steps at fixed θ , iterating $E(z) \leftarrow \sum_m w_m [(z'_m - w'_m) + (1 - \delta)(\kappa + k/q(\theta'_m))]$, where the sum runs over

Gauss–Hermite nodes z'_m with weights w_m and θ'_m is the interpolated policy at z'_m .

5. **Damped update and convergence.** Compute $\text{err} = \max_i |\Delta E_i|$ and update $E \leftarrow \omega E^{\text{new}} + (1 - \omega) E$ with the adaptive ω introduced below. Return to step 2 until $\text{err} < 10^{-7}$ or 2000 passes.

Working from the free-entry condition inherits the fast convergence of policy iteration and never iterates a value function.

Table C.1: Numerical implementation.

Setting	Value
Grid points N ; GH quadrature nodes	25; 7
Productivity AR(1) (ρ_z, σ_z) , monthly	(0.911, 0.0053)
Discount β ; separation δ	$e^{-0.04/12}$; 0.035
Fixed-cost share ϕ	0.90
Tightness bounds $[\theta_{\text{lo}}, \theta_{\text{hi}}]$	$[10^{-3}, 30]$
Howard steps K	15
Outer damping ω	init 0.3, range $[0.02, 0.9]$
Convergence tolerance; max passes	10^{-7} ; 2000
Homotopy on $\sigma(u)/\sigma(z)$	$5.0 \rightarrow 10.8$, 6 steps
Inner solver (b, k)	trust region, ftol 10^{-4} , ≤ 300 iter
Outer search α_L	false position, $[0.05, 0.30]$, tol 0.008, ≤ 6
Simulation length	200,000 months
Burn-in; seed	1,200; 123

Appendix C.2. The bounded- q corner

The cap $q \leq \bar{q}$ makes the inverse q^{-1} steep near the shutdown threshold $\bar{J} \equiv \kappa + k/\bar{q}$: in the near-Leontief left regime q is nearly flat, so a vanishing change in J moves θ sharply away from zero, and an undamped iteration oscillates in a limit cycle at the boundary. Two devices tame this while altering the iteration path but not the fixed point. First, *during* the iteration the hard shutdown is replaced by a narrow smooth ramp: θ is scaled linearly from 0 to its full value over the band $[\bar{J}, \bar{J} + \delta_s]$, with $\delta_s = \text{clamp}(0.15(J_{\text{ss}} - \bar{J}), 10^{-4}, 2k/\bar{q})$ a small fraction of the (narrow) gap between the steady-state J and $\bar{J}$, so the steady state is

never attenuated; the reported policy uses the exact shutdown $\theta = 0$ for $J \leq \bar{J}$. Second, the outer damping weight is adaptive: ω starts at 0.3 and is scaled by 1.1 toward 0.9 when err falls, or by 0.5 toward a floor of 0.02 when it rises. Convergence is judged in absolute terms, $\max_i |\Delta J_i|$, because the thin surplus at the calibration drives J toward zero in slack states, where a relative criterion is unreliable.

Appendix C.3. Estimation and homotopy

For a candidate (b, α_L, k) we fix κ from the fixed-cost share $\phi = 0.90$, solve the model as above, simulate a long panel from the continuous AR(1) process using the solved policy (with interpolated $\theta(z)$ at off-grid z values), and compute the target moments on the resulting stationary distribution as HP-filtered proportional deviations from the mean. The three data moments are matched in a nested loop that exploits their mapping to parameters (Section 6.3):

- an *outer* bracketed (false-position) search chooses α_L over $[0.05, 0.30]$ so that the simulated wage elasticity equals its target, to a tolerance of 0.008 in at most six evaluations; and
- for each α_L , an *inner* trust-region solve chooses (b, k) —reparametrized as $b \in [0.25, 0.995]$ (logistic) and $k = 0.005 + e^x$ —so that simulated $\sigma(u)/\sigma(z)$ and $\bar{\theta}$ hit their targets, with function tolerance 10^{-4} and at most 300 iterations.

The inner solve is reached by *homotopy*: the unemployment-volatility target $\sigma(u)/\sigma(z)$ is raised from 5.0 to its data value 10.8 in six equal steps, and the trust-region solve at each step is warm-started from the previous step’s (b, k) . Continuation is needed because the high-volatility target is a thin-surplus regime in which the job-creation operator is stiff—its contraction modulus approaches one and many states sit near the shutdown corner—so a cold solve at the final target rarely converges. Both calibration and reported moments use a 200,000-month simulation (burn-in 1,200, seed 123), so the calibrated and reported $\sigma(u)/\sigma(z)$ are computed on the same draw. The productivity innovations are held fixed

across evaluations, so the objective is smooth in the parameters. Increasing the Gauss–Hermite node count from 7 to 15 changes the reported moments by less than 1%; we use 7 throughout.

Appendix C.4. Why a global solution

A perturbation or log-linear approximation around the deterministic steady state is blind to precisely the features this exercise studies. The shutdown corner is non-analytic, and the convex tightness policy in the near-Leontief left regime—the source of the skewed unemployment distribution and the countercyclical Hosios wedge—is a global object that a local expansion smooths away. Matching those moments therefore *requires* the global solution; a perturbation-based method would target objects the model cannot, by construction, produce.

The exercise follows the methodological principle of [Petrosky-Nadeau and Zhang \(2021\)](#), who show that “steady state relations hold poorly in simulations” of globally solved DMP models and therefore calibrate their recruiting-cost parameters to moments of the simulated ergodic distribution rather than to steady-state targets. We adopt the same discipline: the moments we match, and the results we report, are computed on the globally simulated stationary distribution, so the same nonlinear dynamics that generate the skewness, the crisis frequency, and the Hosios-wedge volatility also discipline the parameters. What distinguishes the present application is the separation of identification: the matching-function shape is pinned by the horse-race estimates of Section 5.4, and only the three structural parameters (b, α_L, k) —which the matching data cannot speak to—are estimated inside the model.

Appendix D. Parameter Identification and Robustness Checks

This appendix examines the identification of the four 2P-SMCD parameters $(\theta_c, f_c, p_L, p_R)$ by plotting the profile NLS objective. For each parameter in turn, we fix it on a grid spanning the neighborhood of the NLS estimate, re-optimize over the remaining three, and plot the ratio $\text{RSS}(p_i)/\text{RSS}_{\min}$. A sharply curved profile indicates strong identification; a flat profile indicates that the parameter is not separately pinned by the data. The dashed horizontal line marks the 95% confidence bound $1 + \chi_1^2(0.95)/n$.

Figure D.1 reports the pre-break subsample. The crossover tightness θ_c , the junction level f_c , and the right-branch curvature p_R are sharply identified: all three profiles exhibit steep, roughly quadratic valleys with tight 95% confidence intervals ($\theta_c \in [0.35, 0.43]$, $f_c \in [0.34, 0.37]$, $p_R \in [0.71, 0.95]$). The left-branch curvature p_L is identified from below—the profile rejects $p_L < 13.2$ at the 95% level—but is flat from above: once p_L exceeds approximately 15, additional steepness does not improve fit. This reflects the economic content of the parameter. When p_L is large enough, the vacancy-filling rate rises sharply to its cap $q_{\max}$ in slack markets; further steepening produces a nearly identical function over the observed tightness range. What the data identify is that the left branch must be substantially steeper than the right ($p_L \gg p_R$), confirming the asymmetry that drives the near-Leontief crisis mechanism of Section 6. The exact rate of convergence to the Leontief limit is not separately identified from the level f_c .

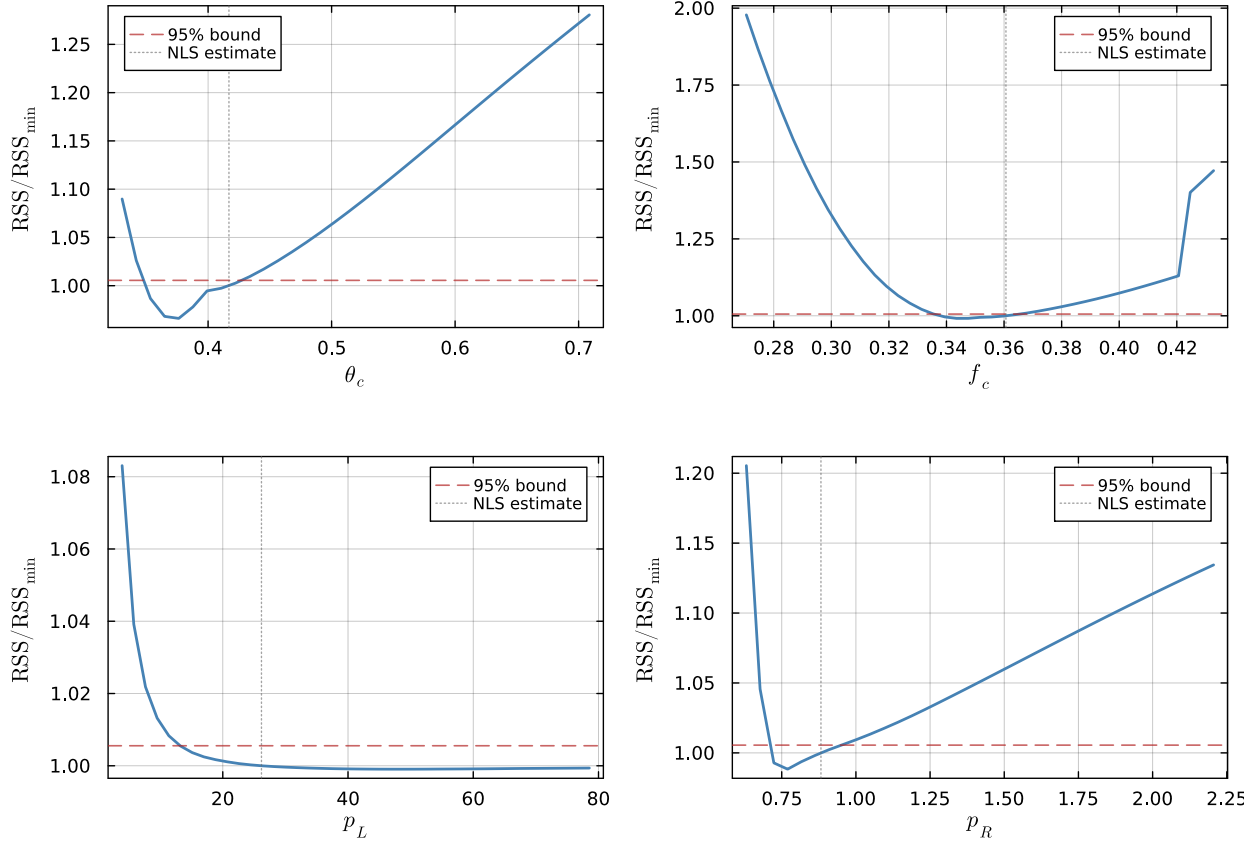


Figure D.1: Profile NLS objective for the 2P-SMCD, pre-break subsample. Each panel fixes one parameter and re-optimizes over the remaining three. Dashed red: 95% confidence bound. Dotted grey: NLS point estimate.

Figure D.2 reports the post-break subsample. The same qualitative pattern holds: θ_c , f_c , and p_R are sharply identified with somewhat wider confidence intervals reflecting the shorter sample ($\theta_c \in [0.25, 0.35]$, $f_c \in [0.20, 0.24]$, $p_R \in [0.95, 1.79]$). The p_L profile again has a sharp lower bound ($p_L \geq 4.7$) and is flat from above. The lower threshold in the post-break sample is consistent with the wider range of tightness observed after 2008—including the COVID trough near $\theta \approx 0.1$ —which provides more information about left-branch behavior and identifies a lower p_L as sufficient for the near-Leontief regime.

The Hessian condition number at the optimum is 1.9×10^5 (pre-break) and 3.7×10^4 (post-break), driven by the flat p_L direction. Excluding p_L , the condition number of the 3×3 sub-Hessian for (θ_c, f_c, p_R) drops to 173 and 205, confirming that the remaining parameters

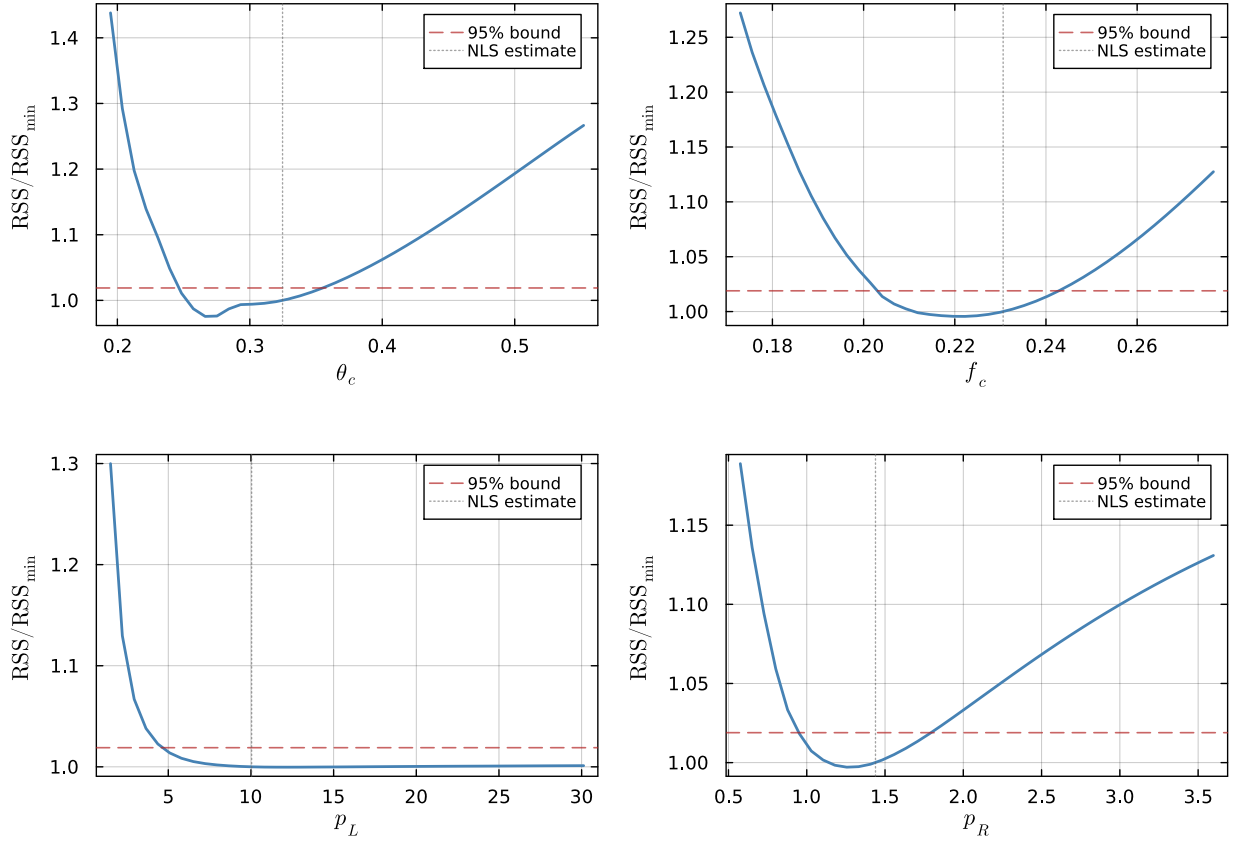


Figure D.2: Profile NLS objective for the 2P-SMCD, post-break subsample. Layout as in Figure D.1.

are sharply pinned.

Appendix D.1. Robustness of the crisis objects to p_L

Because p_L is identified only from below, we verify that the headline quantitative results of Section 6 do not hinge on its exact value. Table D.1 recomputes the crisis objects across the identified pre-break range $p_L \in [13, 40]$, holding the remaining matching-function parameters (θ_c, f_c, p_R) at their pre-break NLS estimates and the structural parameters (b, α_L, k, ϕ) at their calibrated values (Table 7). Only p_L varies; nothing is re-estimated.

Table D.1: Crisis objects across the identified range of p_L (2P-SMCD, pre-break).

p_L	Mean crisis gap (pp)	Skewness of u	Crisis frequency (%)	$\sigma(u)/\sigma(z)$
13.0	12.1	13.3	0.16	9.1
20.0	12.6	14.0	0.20	10.1
26.2 (estimate)	12.9	13.8	0.22	10.6
33.0	13.2	13.9	0.25	11.1
40.0	13.4	13.9	0.27	11.6

Note: Only p_L varies; (θ_c, f_c, p_R) and (b, α_L, k, ϕ) held fixed. Objects from the same 200,000-month global simulation as Section 6. $\sigma(u)/\sigma(z)$ is an untargeted outcome here, since the structural parameters are not recalibrated as p_L moves; it is targeted only at the estimate.

The mean crisis gap stays within a narrow 12.1–13.4 pp band and the skewness of unemployment barely moves (13.3–14.0) across the entire identified range; the vacancy-shutdown mechanism operates (6 productivity states shut down) throughout. The crisis frequency and $\sigma(u)/\sigma(z)$ rise gently with p_L —the latter because the structural parameters are held fixed rather than recalibrated, so the volatility target is met only at the estimate—but neither the sign-reversing gap nor the fat right tail is an artifact of the particular p_L the data fail to pin down from above. The headline results are governed by the *asymmetry* $p_L \gg p_R$, not by the exact left-branch curvature.

Appendix D.2. Sensitivity to the fixed-cost share

The welfare results of Section 6.4 depend on ϕ , the share of total hiring costs attributable to the fixed matching cost κ . The baseline calibration sets $\phi = 0.90$ following [Christiano et al. \(2016\)](#). Because the modified Hosios condition compresses α_L^H toward zero as $\phi \rightarrow 1$, the sign-reversing gap documented in Section 6.4 could be sensitive to this choice. To assess robustness, we recalibrate the entire model at $\phi = 0.80$, holding the three simulation targets $(\sigma(u)/\sigma(z), \bar{\theta}, \varepsilon_{w,y})$ fixed. The three-moment SMM re-estimates (b, α_L, k) : lowering ϕ shifts

the efficient bargaining share upward, so the calibration raises α_L to match the same wage elasticity. It also requires a higher b to generate the same unemployment volatility, because less of the hiring cost is sunk and more responds to congestion.

Table D.2: Sensitivity to fixed-cost share ϕ : calibrated parameters and welfare objects.

		$\phi = 0.80$	$\phi = 0.90$
2P-SMCD	b	0.880	0.769
	α_L	0.125	0.102
	Mean Hosios wedge	−0.107	−0.031
	Mean crisis gap (pp)	+12.6	+12.9
	Skewness of u	13.5	13.8
HRW	b	0.848	0.698
	α_L	0.083	0.061
	Mean Hosios wedge	−0.030	+0.001
	Mean crisis gap (pp)	+4.2	+6.2
	Skewness of u	1.7	1.8

Note: At each ϕ , (b, α_L, k) recalibrated to the same three targets. Mean Hosios wedge = $\bar{\alpha}_L - \bar{\alpha}_L^H$; mean crisis gap = $\bar{u}_{\text{dec}} - \bar{u}_{\text{plan}}$ when $u_{\text{dec}} \geq 15\%$. Skewness: level skewness of the monthly unemployment rate.

Table D.2 reports the results. At $\phi = 0.80$, α_L rises from 0.102 to 0.125 (2P-SMCD) and from 0.061 to 0.083 (HRW), confirming that the low calibrated bargaining weight under $\phi = 0.90$ is driven by the high fixed-cost share rather than an intrinsic feature of the matching function. Matching the same unemployment volatility at lower ϕ requires higher b (0.880 versus 0.769 for the 2P-SMCD; 0.848 versus 0.698 for HRW): with more of the hiring cost responding to congestion, a higher outside option is needed to keep the surplus thin enough for amplification—the [Hagedorn and Manovskii \(2008\)](#) channel, which substitutes

for the fixed-cost channel as ϕ falls. The crisis-episode gap is virtually unchanged (+12.6 versus +12.9 pp under the 2P-SMCD) and unemployment skewness remains an order of magnitude higher than HRW's (13.5 versus 1.7). These objects depend on the matching function's *shape*—specifically p_L and θ_c —and are invariant to the cost decomposition. The mean Hosios wedge widens from -0.031 to -0.107 because α_L^H rises with lower ϕ (from 0.133 to 0.232) while α_L rises less, but the qualitative verdict—countercyclical wedge, sign-reversing unemployment gap, fat-tailed crises under the 2P-SMCD—is robust to the choice of fixed-cost share.

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